Classification of Vector Bundles over an Elliptic Curve

Jack Lipson —— Math 254A —— Fall 2025

University of California, Berkeley

§0. Table of Contents

§1. Abstract

We present M. F. Atiyah's proof of the classification of algebraic vector bundles on elliptic curves defined over algebraically closed fields [Ati57].

§2. Introduction

This expository is intended for students interested in number theory who have little geometry background. Commensurately, we dedicate the most effort to defining what is mentioned in the abstract rather than the classification itself.

This paper increases in abstraction over time. We begin by studying Riemann surfaces from a concrete complex analytical perspective in sections Riemann Surfaces and Elliptic Riemann Surfaces, ultimately classifying the function fields on elliptic Riemann surfaces. This provides great intuition since elliptic Riemann surfaces serve as the prototypical elliptic curves – the curves over $\mathbb{C}$.

That said, these two sections are immediately superseded by sections Algebraic Curves and Elliptic Curves which define and, in some sense, 'classify' elliptic algebraic curves.

The section Vector Bundles defines 'algebraic bundles.' And the section Sheaves practices using the structure sheaf, ultimately proving that finite rank locally free sheaves and finite rank bundles over a variety are equivalent. Line Bundles uses this equivalence to build the Picard group and define the degree of a bundle. Riemann-Roch concludes this sequence by defining sheaf cohomology and stating Riemann-Roch / Serre duality.

Penultimately, Bundle Properties lays out all the lemmas used in the prized Bundle classification.

§3. Riemann Surfaces

Acknowledgment. *This section uses [Don11] and [Tel98] as key references.*

Riemann surfaces are complex curves, or one-dimensional complex manifolds. That is:

Definition 3.1 (Riemann surface). *A Riemann surface X is a Hausdorff topological space covered by open sets $\{U_\alpha\}_\alpha$ where each U_α is equipped with a homeomorphism $\psi_\alpha : U_\alpha \to \tilde{U}_\alpha \subseteq \mathbb{C}$ such that*

$$\psi_\alpha \circ \psi_\beta^{-1} : \psi_\beta(U_\alpha \cap U_\beta) \to \psi_\alpha(U_\alpha \cap U_\beta)$$

is holomorphic for all pairs α, β. This is also called a holomorphic atlas for X.

It is clear from this definition that we can assume $U_\alpha \cong \tilde{U}_\alpha = \Delta$ the open unit disc in $\mathbb{C}$.

Remark 3.2. *Notice we omitted the usual condition on (complex) manifolds requiring second countability (i.e. a countable basis). Surprisingly, Tibor Radó proved in 1925 that this condition is a necessary consequence of our definition [Lim20]. This is the first of many miracles demonstrating how constrained Riemann surfaces truly are.*

Now we define the morphisms.

Definition 3.3 (Holomorphic map). *Given surfaces $(X, U_\alpha, \psi_\alpha)$ and (Y, V_β, ϕ_β), a map $f : X \to Y$ is holomorphic if $\phi_\beta \circ f \circ \psi_\alpha^{-1}$ is holomorphic on a suitable domain for every pair (α, β).*

Here are some examples of surfaces:

Example 3.4.

- *The complex line $\mathbb{C}$ is a Riemann surface with the cover $\mathbb{C}$, $\mathrm{id}_{\mathbb{C}}$.*
- *The projective line $\mathbb{CP}^1 = \{[x : y] : (x, y) \in \mathbb{C}^2 - \{0\}\}$ is also a Riemann surface. Observe $U_0 = \{[1 : \frac{y}{x}]\}$ and $U_\infty = \{[\frac{x}{y} : 1]\}$ are homeomorphic to $\mathbb{C}$ under the obvious projections, $U_0 \cup U_\infty = \mathbb{CP}^1$, and there exists a biholomorphism $\mathbb{C}^\times \to U_0 \to U_\infty \to \mathbb{C}^\times$ via*

$$y \mapsto [1 : y \neq 0] \mapsto [\frac{1}{y} : 1] \mapsto \frac{1}{y}.$$

- *We will soon study perturbations of the elliptic curve $\mathbb{C}/\mathbb{Z}^2$ with obvious charts. This idea generalizes to other 'nice' (free, proper, discontinuous) group actions on $\mathbb{C}$.*

The theory of Riemann surfaces came about largely to resolve multi-valued functions, e.g.

$$f(z) = z^{1/2} = \pm\sqrt{z}.$$

The graph of this function $X = \{(z, w) : w = f(z)\} = \{(re^{i\theta}, re^{i\theta/2})\} \sqcup \{(re^{i\theta}, re^{i(\theta/2+\pi)}\}$ can be visualized as two disjoint copies, or 'sheets,' of $\mathbb{C}$ identified at $z = 0$. The compactification $X^+ = X \cup \{(\infty, \infty)\}$ shares this extra point across both sheets. This is an example of a singular, or non-analytic, Riemann surface. We will largely ignore such surfaces to stay on topic, but thinking of Riemann surfaces as graphs will sometimes come in handy.

Remark 3.5. *Much of the rigidity proven in elementary complex analysis carries over to Riemann surfaces. For instance, none of the following 3 annuli (which themselves are surfaces) are biholomorphic despite obvious homeomorphisms:*

$$\text{(i) } \{0 < |z| < 1\}, \quad \text{(ii) } \{1 < |z| < 2\}, \quad \text{(iii) } \{0 < |z| < \infty\}.$$

We develop some of these strong properties of holomorphic maps.

Lemma 3.6. *Say $f(0) = 0, f \not\equiv 0$ is holomorphic on some neighborhood U of 0. Then there is a unique integer $k \geq 1$ such that $g'(0) \neq 0$ and $f(z) = g(z)^k$ on some $\{0\} \subset V \subseteq U$ for g holomorphic on U.*

Proof. We can write $f(z) = \sum_k^\infty a_i z^i$ for $a_k \neq 0$ and $k \geq 1$ (since we assume $a_0 = 0$). Set

$$g(z) = a_k^{1/k} z \cdot (1 + \frac{a_{k+1}}{a_k} z + \frac{a_{k+2}}{a_k} z^2 + \cdots)^{1/k}.$$

$|\sum_0^\infty \frac{a_{k+i}}{a_k} z^i| = \frac{|f(z)|}{|a_k|} < 1$ on some $V \subseteq U$ since $f(0) = 0$ and f is continuous so the k-th root of this function is holomorphic and well-defined on V. (Note that we do have to make choices for all the k-th roots.) Clearly, $g'(0) = 0$. The uniqueness of k is obvious. □

Proposition 3.7. *Say $f : X \to Y$ is a non-constant holomorphic map between connected Riemann surfaces. Then, for each $x \in X$, there is a unique $k_x \in \mathbb{Z}_{\geq 1}$ such that, in charts around $x, f(x)$, f looks like $z \mapsto z^{k_x}$.*

Definition 3.8 (Valency). *k_x is called the valency of f at x. The last corollary suggests k_x is the (limit of the) number of solutions to $f(z) = w$ for w increasingly near to $f(x)$.*

In other words, f is locally injective at x if and only if $k_x = 1$.

Corollary 3.9 (Open-mapping theorem). *If $f : X \to Y$ is a non-constant holomorphic map between connected Riemann surfaces, then f is an open map.*

It follows that if X is compact, then f is surjective.

Corollary 3.10 (Inverse function theorem). *If $f : X \to Y$ is an injective (so non-constant) holomorphic map between connected Riemann surfaces, then X is isomorphic to $f(X)$.*

In particular, f^{-1} (restricted to a small neighborhood) is analytic wherever f is injective.

Lemma 3.11. *If $f : X \to Y$ is a holomorphic map between connected Riemann surfaces and locally constant at some $x_0 \in X$, then $f(X) = f(x_0)$.*

Proof. Let $C = f^{-1}(x_0)$. C is a closed subset of X. If $C \neq X$, consider a point x on the boundary of C. f is not constant near x so for z in shrinking neighborhoods around x, the number of solutions to $f(z) = f(x_0)$ should tend towards k_x, but of course, any neighborhood of x contains infinitely many points of C - contradiction. So $C = X$; f is constant. ☐

Corollary 3.12. *If $f : X \to Y$ is a non-constant map for connected Riemann surfaces, then the set of points $\{x : k_x > 1\}$ is discrete.*

Proof. Suppose x_0 was an accumulation point of this set. Local to x_0, f' vanishes on this set so locally vanishes by the identity theorem so is 0 on X by our lemma. Since $f' \equiv 0$, all higher derivatives vanish so f is constant - contradiction. ☐

All of this motivates a useful new word:

Definition 3.13 (Degree). *If $f : X \to Y$ is a non-constant holomorphic map between connected, compact Riemann surfaces, then the degree of f*

$$\deg(f) \coloneqq \sum_{x \in f^{-1}(y)} k_x$$

is independent of the choice of $y \in Y$ and finite.

As an aside, we only defined k_x for non-constant maps so if f is constant, we define $\deg(f) = 0$.

Proof. Suppose for contradiction that $\sum_{x \in f^{-1}(y)} k_x = \infty$ for some $y \in Y$. This forces $|f^{-1}(y)| = \infty$. If $f^{-1}(y)$ was discrete, compactness would imply finiteness, so there must exist an accumulation point z of $f^{-1}(y)$ (that is in $f^{-1}(y)$ by continuity of f). Similar to our earlier proofs, $k_z \notin \mathbb{Z}$ so does not exist so f is constant - contradiction. Thus, $\sum_{x \in f^{-1}(y)} k_x < \infty$ for all $y \in Y$ and importantly (since $k_x \geq 1$) $|f^{-1}(y)| < \infty$.

Now, we show that $\deg : Y \to \mathbb{Z}$ is locally constant. Fix some $y \in Y$ so $f^{-1}(y) = \{x_1, \ldots, x_n\}$. Fix open U_i around x_i and open V_i around y so locally $z \mapsto z^{k_{x_i}}$. Put $U = \cup U_i$. $X - U$ is closed so compact so $f(X - U)$ is compact in Y Hausdorff so closed. However, by open-mapping theorem, $f(X) = Y$ is compact so we can find V containing y such that V does not meet $f(X - U)$, i.e. $f^{-1}(V) \subseteq U$. $\tilde{V} = V \cap (\cap_i V_i)$ is an open neighborhood of y on which all local forms apply and such that preimages of any $y' \in \tilde{V}$ will be contained in U. Hence, $\sum_{x \in f^{-1}(y)} k_x = \sum_{x' \in f^{-1}(y')} k_{x'}$ and degree is locally constant. Since this function is integer valued on a connected set, the degree is constant over Y so well-defined. ☐

To play with degree, we need one more word.

Definition 3.14 (Meromorphic map). *A meromorphic function on a Riemann surface X is a holomorphic map $f : X \to \mathbb{CP}^1 = \mathbb{C} \cup \{\infty\}$ not identically ∞.*

Locally, this agrees exactly with meromorphic functions from basic complex analysis, i.e. f will be meromorphic in charts.

Remark 3.15. *The ring of meromorphic maps forms a field.*

We define the order of the pole (i.e. wherever $f(x) = \infty$) to be the valency of x.

Corollary 3.16. *Say X is a compact connected Riemann surface and f is a meromorphic function on X with exactly one pole of order 1.*

Then, $\deg f = \sum_{x \in f^{-1}(\infty)} k_x = k_{f^{-1}(\infty)} = 1$ so f is globally injective but we know that f is surjective since X is compact connected so X is isomorphic to the Riemann sphere.

There is much more we can say about meromorphic maps on $\mathbb{CP}^1$ in fact.

Proposition 3.17. *A non-constant meromorphic function f on $\mathbb{CP}^1$ is uniquely expressible as*

$$p(z) + \sum_{i,j} \frac{a_{i,j}}{(z-p_i)^j}$$

for poles p_i and a polynomial p.

Proof. Since f cannot be identically ∞ by definition of meromorphic, $f^{-1}(\infty)$ must be discrete and thus finite. So we consider poles $p_1, \ldots, p_N$. Local to p_i, we expand f as

$$a_{i,n_i}(z-p_i)^{-n_i} + a_{i,n_i-1}(z-p_i)^{n_i-1} + \cdots + a_{i,1}(z-p_i)^{-1} + \sum_{k \geq 0} a_{i,-k}(z-p_i)^k.$$

$f - \sum_{i=1}^N [a_{i,n_i}(z-p_i)^{-n_i} + a_{i,n_i-1}(z-p_i)^{n_i-1} + \cdots + a_{i,1}(z-p_i)^{-1}]$ only has a pole at ∞ so is meromorphic in the traditional sense and thus a polynomial. $\square$

Proposition 3.18. *A non-constant meromorphic function f on $\mathbb{CP}^1$ is uniquely expressible, given zeroes z_i and poles p_j as*

$$c \times \frac{\prod(z-z_i)}{\prod(z-p_j)}.$$

Proof. $f \cdot \frac{\prod(z-p_j)}{\prod(z-z_i)}$ has no poles so by the last proposition, is just a polynomial. But it is a polynomial with no roots so is constant, i.e. it is $c \in \mathbb{C}^\times$. $\square$

The following result is evident.

Theorem 3.19. *The field of meromorphic functions on $\mathbb{CP}^1$ is precisely the field of rational functions $\mathbb{C}(z)$.*

Observe that, if there are more zeroes than poles in $\mathbb{C}$, there are that many more poles at ∞. And if there are more poles than zeroes in $\mathbb{C}$, there are that many more zeroes at ∞. Hence, the number of zeroes and poles on $\mathbb{CP}^1$

Proposition 3.20. *The degree of a non-constant holomorphic map $f : \mathbb{CP}^1 \to \mathbb{CP}^1$ equals its degree as a rational function.*

Proof. This follows easily from our second unique presentation theorem in that $\deg \frac{p(z)}{q(z)} := \max(\deg p, \deg q)$ for polynomials $p, q \in \mathbb{C}[z]$. $\square$

§4. Elliptic Riemann Surfaces

Acknowledgment. *This section also uses [Don11] and [Tel98] as key references.*

Most of the last section covered the theory of genus 0 Riemann surfaces, like $\mathbb{CP}^1$. To explore higher genus surfaces like elliptic surfaces, we need to actually define genus.

Definition 4.1 (Genus). *A surface X homeomorphic to $\#^g(S^1 \times S^1)$ has genus g.*

Theorem 4.2 (Uniformization). *Up to biholomorphism, there are only three simply connected Riemann surfaces:*

(i) Δ, (ii) $\mathbb{C}$, (iii) $\mathbb{CP}^1$.

Remark 4.3. *Consider a genus one Riemann surface X with universal cover $\tilde{X}$. By the Uniformization Theorem, $\tilde{X} \cong \mathbb{C}$ (because neither $\Delta, \mathbb{CP}^1$ can cover X). Since $\pi_1(X) \cong \mathbb{Z}^2$ by our definition of genus, $X = \tilde{X}/\pi_1(X) \cong \mathbb{C}/\Lambda$ for a lattice Λ.*

We have a special name for these genus one surfaces.

Definition 4.4 (Elliptic Riemann surface). *An elliptic Riemann surface X is a genus one Riemann surface. By the last remark, any such surface is biholomorphic to $\mathbb{C}/\Lambda$ where*

$$\Lambda = \mathbb{Z}\omega_1 \oplus \mathbb{Z}\omega_2, \quad \omega_1^{-1}\omega_2 \notin \mathbb{R}, \quad \omega_i \in \mathbb{C}^\times \quad (\textit{i.e. } \mathbb{R}\textit{-linearly independent})$$

is a free abelian subgroup of $\mathbb{C}$.

Elliptic surfaces are so named because they come from elliptic functions:

Definition 4.5 (Elliptic function). *An elliptic function $f : \mathbb{C} \to \mathbb{P}^1$ is a doubly periodic meromorphic function, i.e. such that*

$$f(z) = f(z + \omega_i), \quad i = 1, 2.$$

It will behoove us to consider the parallelogram with corners $0, \omega_1, \omega_2, \omega_1 + \omega_2$ as a fundamental domain.

Remark 4.6. *We defined elliptic functions as meromorphic because if an elliptic function f is holomorphic, then it is entire on $\mathbb{C}$ and bounded (by the compact image of the parallelogram) so constant per Liouville's theorem.*

Theorem 4.7. *Consider a non-constant elliptic function f with zeroes $z_1, \dots z_n$ and poles $p_1, \dots, p_m$ in the period parallelogram, with repeats for multiplicity. Then:*

$$\text{(i) } m = n, \quad \text{(ii) } \sum_1^n \operatorname{Res}_{p_i} f = 0, \quad \text{(iii) } \sum_1^n z_i = \sum_1^n p_i \mod \Lambda.$$

Proof. (i) From basic complex analysis,

$$\oint_{\varparallelogram} \frac{f'(z)}{f(z)} dz = n - m.$$

However, the left and right hand sides of $\varparallelogram$ are the same, being traversed in opposite directions. The same goes for the top and bottom. Hence the integral is 0 so $n = m$.

(ii)

$$\sum_1^n \operatorname{Res}_{p_i} f = \frac{1}{2\pi i} \oint_{\varparallelogram} f = 0.$$

(iii)

$$\sum z_i - \sum p_i = \frac{1}{2\pi i} \oint_{\varparallelogram} z \frac{f'(z)}{f(z)} dz.$$

This time, because of z in the integrand, the values along opposite sides differ. But, still:

$$\begin{aligned}
\oint_{\varparallelogram} z\frac{f'(z)}{f(z)}dz &= \int_0^{\omega_1} z\frac{f'(z)}{f(z)}dz + \int_{\omega_1}^{\omega_1+\omega_2} z\frac{f'(z)}{f(z)}dz + \int_{\omega_1+\omega_2}^{\omega_2} z\frac{f'(z)}{f(z)}dz + \int_{\omega_2}^{0} z\frac{f'(z)}{f(z)}dz \\
&= \int_0^{\omega_1} z\frac{f'(z)}{f(z)}dz + \int_0^{\omega_2} (\omega_1 + z)\frac{f'(z)}{f(z)}dz - \int_0^{\omega_1} (z + \omega_2)\frac{f'(z)}{f(z)}dz - \int_0^{\omega_2} z\frac{f'(z)}{f(z)}dz \\
&= -\int_0^{\omega_1} \omega_2\frac{f'(z)}{f(z)}dz + \int_0^{\omega_2} \omega_1\frac{f'(z)}{f(z)}dz \in 2\pi i(-\mathbb{Z}\omega_2 + \mathbb{Z}\omega_1) \equiv 0 \mod \Lambda.
\end{aligned}$$

□

With these facts, we start building a very specific elliptic function.

Definition 4.8 (Weierstraß $\wp$-function). *The* Weierstraß *$\wp$-function is defined by:*

$$\wp(z) = \frac{1}{z^2} + \sum_{w \in \Lambda^*} \left[\frac{1}{(z-w)^2} - \frac{1}{w^2} \right].$$

First, let us show this series even converges.

Proof. Fix a compact $K \subseteq \mathbb{C}$. We can find

$$a > \sup_{z \in K,\, w \in \Lambda - K} \frac{|w|}{|w-z|}, \quad b > \sup_{z \in K} |z|.$$

Hence, for $z \in K, w \in \Lambda - K$:

$$\left| \frac{1}{(z-w)^2} - \frac{1}{w^2} \right| < \frac{|z|^2 + 2|z||w|}{|w|^2 |z-w|^2} < \frac{a^2 b^2}{|w|^4} + 2\frac{a^2 b^3}{|w|^3}.$$

Integrating $\int_{|w|>1} |w|^{-3} dxdy$ shows the sum, ignoring the finitely many poles of $\Lambda \cap K$, will converge absolutely. $\square$

Observe $\wp$ is even whereas $\wp'$ is odd, as can be seen from

$$\wp'(z) = -\sum_{w \in \Lambda} \frac{2}{(z-w)^3}.$$

Remark 4.9. *Moreover, it is clear that*

$$0 = \wp'(z) - \wp'(z + \omega_i) = [\wp(z) - \wp(z + \omega_i)]'$$

so $\wp(z) - \wp(z+\omega_i)$ is constant and thus equal to $\wp(-\omega_i/2) - \wp(\omega_i/2)$ but this quantity vanishes since $\wp$ is even. Thus, $\wp(z)$ is Λ-periodic. $\wp'$ is immediately recognizable as Λ-periodic from the expression above.

Indeed, $\wp, \wp'$ are elliptic functions on $X = \mathbb{C}/\Lambda$! Also notice $\wp$ is of degree 2 and $\wp'$ is of degree 3 (from checking the order/valency of the single poles in $\varnothing$).

Proposition 4.10. *If we define*

$$G_r = \sum_{w \in \Lambda^*} \frac{1}{w^r}, \quad g_2 = 60 G_4, \quad g_3 = 140 G_6,$$

then

$$(\wp')^2 = 4\wp^3 - g_2 \wp - g_3.$$

Proof. We construct the Laurent expansion for $\wp$ centered at 0. Taking the derivative of $\frac{1}{1-x} = 1 + x + x^2 + \cdots$ gives $\frac{1}{(1-x)^2} = 1 + 2x + 3x^2 + \cdots$ so for $w \neq 0$:

$$\frac{1}{(z-w)^2} - \frac{1}{w^2} = \frac{1}{w^2} \frac{1}{(1 - z/w)^2} - \frac{1}{w^2} = \frac{1}{w^2}(2\frac{z}{w} + 3\frac{z^2}{w^2} + \cdots).$$

$\wp$ is an even function so we can ignore all odd powers of z. Thus,

$$\wp(z) = \frac{1}{z^2} + 0 + \left(\sum_{w \in \Lambda^*} \frac{3}{w^4} \right) z^2 + \left(\sum_{w \in \Lambda^*} \frac{5}{w^6} \right) z^4 + \cdots$$

It follows that

$$\wp'(z) = -\frac{2}{z^3} + \left(\sum_{w \in \Lambda^*} \frac{6}{w^4} \right) z + \left(\sum_{w \in \Lambda^*} \frac{20}{w^6} \right) z^3 + \cdots$$

So

$$(\wp'(z))^2 = \frac{4}{z^6} + 2\cdot(-2)\cdot\left(\sum_{w\in\Lambda^*}\frac{6}{w^4}\right)\frac{1}{z^2} + 2\cdot(-2)\cdot\left(\sum_{w\in\Lambda^*}\frac{20}{w^6}\right) + \cdots$$

And moreover,

$$\wp^3(z) = \frac{1}{z^6} + 3\left(\sum_{w\in\Lambda^*}\frac{3}{w^4}\right)\frac{1}{z^2} + 3\left(\sum_{w\in\Lambda^*}\frac{5}{w^6}\right) + \cdots$$

Therefore,

$$4\wp^3(z) - 60\left(\sum_{w\in\Lambda^*}\frac{1}{w^4}\right)\wp(z) - 140\left(\sum_{w\in\Lambda^*}\frac{1}{w^6}\right) + z^2 f(z) = (\wp'(z))^2 \quad \text{for } f \text{ holomorphic.}$$

Since $z^2 f(z)$ is elliptic and holomorphic, it is constant. This forces $f = 0$. □

Remark 4.11. *We will soon see that (almost) all elliptic curves take the form $y^2 = x^3 + ax + b$, of which the last proposition is very reminiscent.*

Definition 4.12 (Branch point). *$z_0 \in \mathbb{P}^1$ is a branch point meromorphic function f on a Riemann surface X if its inverse image contains points of valency strictly greater than 1.*

Proposition 4.13. *$\wp : \mathbb{C}/\Lambda \to \mathbb{P}^1$ is a degree 2 holomorphic map with branch points over distinct values*

$$e_1 = \wp(\frac{\omega_1}{2}), \quad e_2 = \wp(\frac{\omega_2}{2}), \quad e_3 = \wp(\frac{\omega_1+\omega_2}{2}), \quad \infty.$$

Proof. We found the Laurent expansion for $\wp$ so it is meromorphic. $\wp$ is degree 2 by the order of its poles at Λ. It follows from Theorem 4.7 that $\wp(z) - a$ has exactly two zeroes with repeats for multiplicity for any $a \in \mathbb{C}$. Because double roots correspond to zeroes the derivative, and since $\deg \wp' = 3$ (also per its Laurent expansion), there are at most 3 such a where $\wp - a$ has a double root on $\mathbb{C}/\Lambda$.

e_i are these roots because $\wp'(\omega_i/2) = \wp'(-\omega_i/2) = -\wp'(\omega_i/2)$ and $\wp'(\frac{\omega_1+\omega_2}{2}) = \wp'(-\frac{\omega_1+\omega_2}{2}) = -\wp'(\frac{\omega_1+\omega_2}{2})$ by Λ-periodicity and the fact $\wp'$ is odd. The e_i are distinct; otherwise $\wp - e_i$ would have 4 zeroes (seeing that the ω_i are $\mathbb{R}$-linearly independent). □

We need a lemma.

Lemma 4.14. *Say $f : X \to Y$ is a continuous map between Riemann surfaces that is holomorphic except at isolated points. Then f is holomorphic everywhere.*

Proof. Say $x_0 \in X$ is such an isolated point. Fixing charts containing no other bad points, we can reduce this problem to $f : \Delta \to \mathbb{C}$ continuous everywhere and holomorphic on $\Delta^\times$. Riemann's removable singularity theorem implies f is holomorphic on Δ. □

Proposition 4.15. *Every elliptic function can be decomposed into rational functions of $\wp, \wp'$:*

$$\overbrace{R_1(\wp)}^{\text{even}} \quad + \quad \overbrace{\wp' \cdot R_2(\wp)}^{\text{odd}}, \quad R_1, R_2 \in \mathbb{C}(z).$$

Proof. The above proposition implies $(\mathbb{C}/\Lambda)/(\mathbb{Z}/2\mathbb{Z}) \overset{\wp}{\cong} \mathbb{CP}^1$ is a homeomorphism. $\wp$ is a biholomorphism away from the points of valency greater than 1, but these are exactly $\omega_1/2, \omega_2/2, (\omega_1+\omega_2)/2$. Such points are isolated so we conclude $\wp$ is a biholomorphism everywhere by the previous lemma.

Now consider any even holomorphic function $f : \mathbb{C}/\Lambda \to \mathbb{CP}^1$. Pre-composing with $\wp$, this descends to a holomorphic map $\overline{f} : \mathbb{CP}^1 \to \mathbb{CP}^1$, i.e. $f = \overline{f} \circ \wp$. Theorem 3.19 implies $\overline{f}$ is a rational function so f is a rational function of $\wp$. For an odd holomorphic function $f : \mathbb{C}/\Lambda \to \mathbb{CP}^1$, $f/\wp'$ is even, hence a rational function of $\wp$.

$f = \frac{f(z)+f(-z)}{2} + \frac{f(z)-f(-z)}{2}$ is a decomposition into even and odd parts so we are done. □

We have successfully classified the meromorphic functions on $\mathbb{C}/\Lambda$.

Theorem 4.16. *The field of meromorphic functions on $\mathbb{C}/\Lambda$ is isomorphic to*

$$\mathbb{C}(z)[w]/(w^2 - 4z^3 + g_2 z + g_3),$$

where g_2, g_3 are determined by Λ.

§5. Algebraic Curves

Acknowledgment. *This section primarily references [Sil86].*

Now, we explore algebraic elliptic curves over an algebraically closed field k. It is valuable in number theory to consider the 'rational points' defined over a subfield of k, but we excise this theory since the later classification will not use it.

Definition 5.1 (Affine n-space). *Affine n-space over a field k is*

$$\mathbb{A}^n = \mathbb{A}^n(k) = \{(x_1, \dots, x_n) : x_i \in k\}.$$

Definition 5.2 (Affine algebraic sets). *Consider an ideal $I \lhd k[x] = k[x_1, \dots, x_n]$. Define*

$$V(I) = \{P \in \mathbb{A}^n : f(x) = 0 \text{ for all } f \in I\}.$$

An (affine) algebraic set V is any set of the form $V(I)$ for some ideal I.

By the following theorem, we may write any $I = (f_1, \dots, f_j)$ for some j.

Theorem 5.3 (Hilbert basis). *Ideals of $k[x]$ are finitely-generated for any Noetherian ring k,.*

Definition 5.4 (Affine variety). *If V is an affine algebraic set, the ideal of V is*

$$I(V) = \{f \in k[x] : f(P) = 0 \text{ for all } P \in V\}.$$

V is an affine variety if $I(V)$ is prime in $k[x]$.

Definition 5.5 (Affine coordinate ring, function field). *If V is an affine variety, then the integral domain*

$$k[V] = k[x]/I(V)$$

is called the affine coordinate ring of V. $k(V) := \mathrm{Frac}(k[V])$ is the function field of V.

Definition 5.6 (Dimension). *The dimension $\dim V$ is the transcendence degree of $k(V)/k$.*

A commutative algebra result following from Noether's normalization lemma implies:

Theorem 5.7 ([AM94]). $\dim k[V] = \dim V$.

Here are some examples of (non)-varieties:

Example 5.8. *Put $k = \mathbb{C}$.*

- *The affine algebraic set $\mathbb{A}^n = V(0)$ is actually an affine variety because $I(\mathbb{A}^n) = 0$ is a prime ideal of $\mathbb{C}[x]$. Notice $\mathbb{C}[\mathbb{A}^n] = \mathbb{C}[x]/0 = \mathbb{C}[x_1, \dots, x_n]$ so $\mathbb{C}(\mathbb{A}^n) = \mathbb{C}(x_1, \dots, x_n)$ is of transcendence degree n over $\mathbb{C}$, so $\dim \mathbb{A}^n = n$.*
- *Consider $I = (x-2)^2 \subset \mathbb{C}[x]$. Then $V = V(I) = \{2\}$ and $I(V) = (x-2)\mathbb{C}[x]$. Hence, $\mathbb{C}(V) = \mathbb{C}[x]/(x-2) \cong \mathbb{C}$ is of transcendence degree 0 so the point 2 is dimension 0 as an affine variety.*
- *Consider $I = (xy^3) \subset \mathbb{C}[x,y]$ So $V = V(I) = \{(x,0) : x \in \mathbb{C}\} \cup \{(0,y) : y \in \mathbb{C}\}$. Given $f \in I(V)$, $f(0,y)$ is a polynomial in $\mathbb{C}[y]$. Since $k = \mathbb{C}$, we can factor it completely. But it is identically 0 so any $y^{>0}$ term in f must have a coefficient in $x^{>0}$. This argument is symmetric so $I(V) = (xy)$ which is not prime. Hence V is not an affine variety.*

Definition 5.9 (Non-singular (smooth)). *Say V is an affine variety with point p such that $f_1, \dots, f_k$ generate $I(V) \subset k[x]$. V is non-singular at p if the matrix $\left(\frac{\partial f_i}{\partial x_j}(p)\right)_{i,j}$ has rank $n - \dim V$.*

If V is nonsingular at all $p \in V$, then V itself is non-singular (smooth).

Example 5.10. *Take the affine algebraic set V cut out by $I = (x^2 - y^3) \lhd \mathbb{C}[x, y]$. This is prime but not maximal since it is contained in the ideal (x, y). Hence, V is an affine variety of dimension 1. If $f(x, y) = x^2 - y^3$, then our derivative matrix $\begin{pmatrix} 2x & 3y \end{pmatrix}$ is of rank $2 - 1 = 1$ everywhere except $x = y = 0$.*

So V is not smooth because it is singular at $(0, 0) \in V$.

Definition 5.11 (Cotangent space). *Let the ideal $\mathfrak{m}_p = \{f \in k[V] : f(p) = 0\}$. This is the kernel of $k[V] \overset{ev_p}{\twoheadrightarrow} k$, hence maximal. We define the cotangent space at p:*

$$T_p^* V = \mathfrak{m}_p / \mathfrak{m}_p^2.$$

V is non-singular at p if and only if $\dim_k T_p^* = \dim V$.

Proof. We regurgitate the proof from [Har77].

Put $p = (p_1, \ldots, p_n)$ and $V = V(I)$ for $I = (f_1, \ldots, f_\ell) \subseteq k[x_1, \ldots, x_n]$.

Observe $\mathfrak{m}_p' = (x_1 - p_1, \ldots, x_n - p_n)$ is maximal in $k[x_1, \ldots, x_n]$ so $\mathfrak{m}_p \subset \mathfrak{m}_p'/I$ implies $\mathfrak{m}_p = \mathfrak{m}_p'/I$.

Now consider $T : A \to k^n$ via $f \mapsto (\partial_j f(p))_1^n$. The collection $T(x_i - p_i) = e_i$ form a basis for k^n so $\dim_k \mathfrak{m}_p' \geq n$. However, $\mathfrak{m}_p'/{\mathfrak{m}_p'}^2$ is effectively only linear terms so is spanned by $(x_i - p_i)_1^n$, hence at most dimension n as a k-vector space. Thus $\dim_k \mathfrak{m}_p'/{\mathfrak{m}_p'}^2 = n$.

As $k[X_i]_p/\mathfrak{m}_p$-vector spaces then,

$$\mathfrak{m}_p/\mathfrak{m}_p^2 \cong \mathfrak{m}_p'/({\mathfrak{m}_p'}^2 + I) \quad \text{implying} \quad \dim \mathfrak{m}_p/\mathfrak{m}_p^2 = n - \operatorname{rk}(\partial_j f_i(p))_{ij}.$$

V is smooth at p if and only if the rank of the Jacobian matrix is $n - \dim V$ if and only if $\dim T_p^* = \dim V$. ☐

Definition 5.12 (Local ring, regular). *The local ring of V at p $k[V]_P$ is $k[V]$ localized at $\mathfrak{m}_p$, i.e.*

$$k[V]_p = \left\{ \frac{f}{g} \in k(V) : f, g \in k,\ g(p) \neq 0 \right\}.$$

Functions $f \in k[V]_p$ are said to be regular (or defined) at p.

With these foundations, we add 'points at infinity.'

Definition 5.13 (Homogeneous). *A polynomial $f \in k[x] = k[x_0 \ldots, x_n]$ is homogeneous of degree d if*

$$f(\lambda x_0, \ldots, \lambda x_n) = \lambda^d f(x_0, \ldots, x_n), \qquad \lambda \in k.$$

An ideal $I \lhd k[x]$ is homogeneous if it is generated by homogeneous polynomials.

Definition 5.14 (Projective n-space). *Projective n-space over a field k is*

$$\mathbb{P}^n = \mathbb{P}^n(k) = \{(x_0, \ldots, x_n) \in (\mathbb{A}^{n+1})^*\}/((x_i) \sim (\lambda x_i)) = \{[x_0 : \cdots : x_n] : \textit{not all } x_i = 0\}.$$

Observe that, for f homogeneous, $f(x_0, \ldots, x_n) = 0$ cuts out a subset of $\mathbb{P}^n$, even though $f^{-1}(a \in k^\times)$ does not make sense as a subset of $\mathbb{P}^n$.

Definition 5.15 ((Projective) algebraic set). *For a homogeneous ideal $I \lhd k[x]$, define*

$$V(I) = \{x \in \mathbb{P}^n : f(x) = 0 \textit{ for all } f \in I\}.$$

A (projective) algebraic set V is any set of the for $V(I)$ for some homogeneous ideal I.

Definition 5.16 ((Projective) variety). *Like before, define*

$$I(V) = \{f \in k[x] : f \text{ homogeneous}, f|_V \equiv 0\}.$$

A projective algebraic set V is a <u>*(projective) variety*</u> *if $I(V)$ is prime in $k[x]$.*

Remark 5.17. *We can cover $\mathbb{P}^n$ with copies of affine n-space as follows:*

$$\mathbb{A}^n_i = \{[x_0 : \cdots : x_n] : x_i \neq 0\} \longleftrightarrow \mathbb{A}^n, \quad [x_0 : \cdots : x_n] \longleftrightarrow (\frac{x_0}{x_i}, \ldots, \frac{\hat{x_i}}{x_i}, \ldots, \frac{x_n}{x_i}).$$

Given an affine algebraic set $V \subseteq \mathbb{A}^n$, if we make a choice of i, then $V \subseteq \mathbb{A}^n \cong \mathbb{A}^n_i \subseteq \mathbb{P}^n$. So we can define the <u>*projective closure*</u>

$$\overline{V} = V(I(V)) = V\left(\{f^*(x_0, \ldots, x_n) \coloneqq x_i^{\deg f}(\frac{x_0}{x_i}, \ldots, \frac{\hat{x_i}}{x_i}, \ldots, \frac{x_n}{x_i}) : f \in I(V)\}\right).$$

If V is an affine variety, then $\overline{V}$ is a projective variety. $\overline{V} - V$ are called the points at infinity.

Proposition 5.18. *Conversely, if V is a projective variety, then $V \cap \mathbb{A}^n_i$ is an affine variety.*

Proof. Say $V = V(I)$ for a homogeneous prime ideal $I \lhd k[x_0, \ldots, x_n]$. Then,

$$I(V \cap \mathbb{A}^n_i) = \{f \in k[x_0, \ldots, \hat{x_i}, \ldots, x_n] : f(V \cap \mathbb{A}^n_i) = 0\}.$$

Like in the definition of projective closure, we can homogenize f into a unique element of $I(V)$ by tacking on the minimal number of powers of x_i to each term in f. Hence,

$$f(x_0, \ldots, \hat{x_i}, \ldots, x_n) = f^*(x_0, \ldots, 1, \ldots, x_n).$$

Now suppose $f \cdot g \in I(V \cap \mathbb{A}^n_i)$ for $f, g \in k[(x_{j \neq i})^n_1]$. Thus, $f^* \cdot g^* \in I(V)$ so without loss of generality $f^* \in I(V)$. $f = f^*(x_0, \ldots, \overset{i-th}{1}, \ldots, x_n) \in I(V \cap \mathbb{A}^n_i)$ so $I(V \cap \mathbb{A}^n_i)$ is a prime ideal, making $V \cap \mathbb{A}^n_i$ an affine variety. □

The morphism intended by $\cong$ in the above proof is ambiguous. One way to read it is topological:

Definition 5.19 (Zariski topology). *A subset of $\mathbb{A}^n(\mathbb{P}^n)$ is* <u>*closed*</u> *if it is the common zero-locus $V(I)$ for some (homogeneous) ideal I. This defines the* <u>*Zariski topology*</u>*.*

Notice any $\mathbb{A}^n_i$ is dense in $\mathbb{P}^n$. That is exactly why the following definitions make sense.

Definition 5.20 (Dimension, function field, non-singular, local ring, regular). *Fix a projective variety V. Choose $\mathbb{A}^n_i \subseteq \mathbb{P}^n$ with $p \in V \cap \mathbb{A}^n_i$. Then:*

<u>dimension of V</u>	$=$	$\dim V$	$=$	$\dim(V \cap \mathbb{A}^n_i)$
<u>function field of V</u>	$=$	$k(V)$	$=$	$k(V \cap \mathbb{A}^n_i)$
<u>V non-singular at p</u>			$=$	$V \cap \mathbb{A}^n_i$ non-singular at p
<u>local ring of V at p</u>	$=$	$k[V]_p$	$=$	$k[V \cap \mathbb{A}^n_i]_p$
<u>regular functions of V at p</u>	$=$	$\{f \in k(V)\}$	$=$	$\{f \in k(V \cap \mathbb{A}^n_i)\}$

Proof. The above definitions are canonically isomorphic over choices of i by redefining $k(V)$, i.e.

$$k(V) = \left\{\frac{f}{g} : f, g \in k[x_0, \ldots, x_n] \ \underline{\text{homogeneous}} \text{ and of } \underline{\text{same degree}} \text{ and } \underline{g \notin I(V)}\right\}.$$

The following map is well-defined and bijective (where the inverse is evaluation at $x_i = 1$):

$$k(V \cap \mathbb{A}^n_i) \overset{k-\text{alg}}{\cong} k(V), \quad \frac{f(x_0, \ldots, \widehat{x_i}, \ldots, x_n)}{g(x_0, \ldots, \widehat{x_i}, \ldots, x_n)} \longleftrightarrow \frac{f^*}{g^*} = \frac{x_i^{\max\{\deg f, \deg g\}} f(x_0/x_i, \ldots, x_n/x_i)}{x_i^{\max\{\deg f, \deg g\}} g(x_0/x_i, \ldots, x_n/x_i)}$$

The remaining definitions follow easily. □

Definition 5.21 (Rational map). *A function $\phi : V \to W$ is rational for $V \subseteq \mathbb{A}^m$, $W \subseteq \mathbb{A}^n$ if*

$$\phi = (\phi_1, \dots, \phi_n) \quad \textit{for} \quad \phi_i \in k[V].$$

Definition 5.22 (Regular map). *A function $\phi : V \to W$ is regular for $V \subseteq \mathbb{P}^m$, $W \subseteq \mathbb{P}^n$ if ϕ is continuous with respect to the Zariski topology and locally rational.*

More concretely, $\phi = [\phi_0 : \cdots : \phi_n] : V \to W$ for $\phi_i \in k(V)$ and for every $p \in V$, there exists some $\psi \in k(V)$ so not all $(\psi \cdot \phi_i)(p) = 0$ and each $(\psi \cdot \phi_i) \in k(V)_p$.

This defines the morphisms between affine/projective varieties.

Definition 5.23 (Curve). *A curve is a projective variety of dimension one.*

Proposition 5.24. *If X is a curve and $p \in X$ a non-singular point, then $k[X]_p$ is a discrete valuation ring.*

Proof. Say $p = (p_i)_1^n \in X_i = X \cap \mathbb{A}_i^n = V(I)$ for $(f_1, \dots, f_\ell) = I \lhd k[x_1, \dots, x_n]$. Recall $k[X]_p = k[X_i]_p$ is $k[X_i] = k[x_1, \dots, x_n]/(f_1, \dots, f_\ell)$ localized at $\mathfrak{m}_p = \{g \in k[X_i] : g(p) = 0\}$.

From our definition of dimension through the cotangent space, since X is smooth at p, $\dim \mathfrak{m}_p/\mathfrak{m}_p^2 = \dim X = 1$. $\dim k[X_i]_p \leq \dim k[X_i] = \dim X = 1$ but of course $\mathfrak{m}_p \neq 0$ so $k[X_i]_p$ is a Noetherian local domain of dimension 1. Then theorem 9.2 in [AM94] shows $k[X_i]_p$ is a discrete valuation ring. □

Remark 5.25. *In particular, the valuation on $k[X]_p$ is given by:*

$$ord_p : k[X]_p \to \mathbb{N} \cup \{\infty\}, \quad ord_p(f) = \sup\{d \in \mathbb{Z} : f \in \mathfrak{m}_p^d\}.$$

This extends to $k(X) \to \mathbb{Z} \cup \{\infty\}$ via $ord_p(f/g) = ord_p(f) - ord_p(g)$.

The uniformizer is given by any $\pi \in \mathfrak{m}_p - \mathfrak{m}_p^2$, i.e. $(\pi) = \mathfrak{m}_p$.

Definition 5.26 (Zero, pole). *The order of $f \in k(X)$ at $p \in X$ is $ord_p(f)$. p is a zero (pole) of f if $ord_p(f) > 0$ ($ord_p(f) < 0$). If $ord_p(f) < 0$, we say $f(p) = \infty$.*

Remark 5.27. *Notice f is regular at p if and only if $ord_p(f) \geq 0$.*

Lemma 5.28. *Fix $f \in k(X)^*$. f has finitely many poles and zeroes. If it has neither, $f \in k$.*

Proof. We show f has finitely many zeroes. Then $1/f$ will have finitely many zeroes, i.e. f will have finitely many poles.

Fix $f \notin k$. k is algebraically closed so f is transcendental over k. Since $\dim X = 1$, $k(X)$ is also transcendental of degree 1 over k so $k(X)$ is a finite extension of $k(f)$.

It is well known that the integral closure B of $k[f]$ in $k(X)$ is a Dedekind domain. B has finitely many maximal ideals which are in 1-1 correspondence with points where f vanishes. In particular, B must have at least one maximal so f cannot have no zeroes. □

To continue, we must better understand the maps between curves.

Lemma 5.29. *If X is a projective variety and Y is a variety, then the morphism $\phi : X \to Y$ is a closed map so $\phi(X) = V(I)$ for some ideal I. Moreover, its image is a variety, i.e. I is prime.*

Proof. Say $V \subseteq X \subseteq \mathbb{P}^m$ is closed, i.e. $V = V(p_1, \dots, p_j)$ for homogeneous polynomials $f_i \in k[x_0, \dots, x_m]$. The image of V in $\mathbb{P}^m \times Y$ under $i \times \phi$ is closed since, in open sets $\mathbb{A}_i^m \times Y$, it is cut out by $p_\ell|_{x_i=1}$ and $y_\ell = \phi_\ell$ for Y-coordinates ℓ. It is easy to see, again looking at affines now covering Y, that $\mathbb{P}^n \times Y \twoheadrightarrow Y$ is closed so $\phi(X)$ is an algebraic set.

We claim $I(V(I))$ is prime. If it was not then there would exist $f, g \notin I(V(I))$ with $fg \in I(V(I))$. This implies neither f nor g vanish on all of V yet $\phi(X) = V(I(V(I)), f) \cup V(I(V(I)), g)$.

This is the union of two closed sets so by the continuity of ϕ, the preimage of each subset is closed and makes up X. If both subsets were proper and nonempty, we could take a function from the ideal of each preimage not in the other ideal (i.e. neither vanishing everywhere on X). This product is in $I(X)$, which is prime, so one function is in $I(X)$ but that means it vanishes on X - contradiction. So $\phi(X)$ is a subvariety. $\square$

The next few results should be very reminiscent of our study of Riemann surfaces.

Proposition 5.30. *If X is a smooth curve and Y is a curve, then the morphism $\phi : X \to Y$ is either surjective or constant.*

Proof. Per the lemma, $\phi(X)$ is a projective subvariety in Y. So the inclusion $\phi(X) \hookrightarrow Y$ induces $k[Y] \twoheadrightarrow k[\phi(X)]$ which must be the quotient by a prime ideal containing $I(Y)$.

However, $\dim k[Y] = \dim Y = 1$ implies $0 = I(Y) \subseteq I(X)$ is a chain of at most length 2. So either $I(X) = I(Y)$ in which case ϕ is surjective or $I(X) \neq I(Y)$ is a maximal ideal corresponding to a point on Y in which case ϕ is constant. $\square$

Corollary 5.31. *Suppose ϕ is non-constant.*

If $\phi^*(f) = f \circ \phi = 0$, the surjectivity of ϕ implies $f \equiv 0$ for $f \in k(Y)$ so $\phi^* : k(Y) \hookrightarrow k(X)$. Because $\dim X = \dim Y = 1$, both fields are of transcendence degree 1 over k so $k(X)$ is a finite extension of $k(Y)$.

Definition 5.32 (Degree). *If $\phi : X \to Y$ is a non-constant morphism of curves, then the degree of ϕ is $[k(X) : k(Y)] < \infty$. If ϕ is constant, we say the degree of ϕ is 0.*

Remark 5.33. *Note it is possible to recover a local ring on a smooth curve from any discrete valuation ring of a dimension 1 function field over k (a finitely-generated transcendence degree 1 extension of k). Section I.6 of [Har77] uses this reconstruction to get the following theorem.*

Theorem 5.34. *There is the following equivalence of categories:*

$$\left\{\begin{matrix}\text{objects: smooth curves over } k \\ \text{morphisms: surjective regular maps}\end{matrix}\right\} \longleftrightarrow \left\{\begin{matrix}\text{objects: fin.-gen. transcendence deg 1 extensions of } k \\ \text{morphisms: field inclusions fixing } k\end{matrix}\right\}$$

Corollary 5.35. *A degree one map $\phi : X \to Y$ of smooth curves an isomorphism.*

After all, this paper is an expository on number theory so we might as well mention the following:

Remark 5.36. *A non-constant map ϕ of smooth curves is said to be separable, inseparable, purely inseparable, if its induced field extension has the respective property.*

The ramification index of $\phi : X \to Y$ of $p \in X$ is, for a uniformizer t of $k[Y]_{\phi(p)}$

$$e_\phi(p) = ord_p(\phi^* t) \geq 1.$$

ϕ is unramified at p if $e_\phi(p) = 1$ and unramified if unramified on all of X.

In fact, for any fixed $q \in Y$:

$$\deg \phi = \sum_{p \in \phi^{-1}(q \in Y)} e_\phi(p).$$

The proof of this latest fact is quite similar to the proof of the same definition/theorem for Riemann surfaces, i.e. showing local invariance via a unit at q with local parameter t and then playing with valuations as opposed to valencies.

§6. Elliptic Curves

Acknowledgment. *This section again relied mostly on [Sil86].*

We finally define *elliptic* algebraic curves and prepare to discuss bundles.

Definition 6.1 (Divisor group). *The divisor group of a curve X is the free abelian group*

$$\mathrm{Div}(X) = \left\{ D = \sum_{p \in X \text{ distinct}} n_p p : \text{finitely many } n_p \neq 0 \right\}.$$

By the finiteness of poles and zeroes, the following map is well-defined:

$$\mathrm{div} : k(X)^* \to \mathrm{Div}(X), \quad f \mapsto \sum_{p \in X} \mathrm{ord}_p(f).$$

Definition 6.2 (Divisor class/Picard group). *A divisor D is principal if $D = div(f)$ for some $f \in k(X)^*$. The Picard group of X is*

$$\mathrm{Pic}(X) = \mathrm{Div}(X)/\{\text{principal divisors}\}.$$

Definition 6.3 (Degree). *Given $D = \sum_1^n k_i p_i \in Div(X)$, the degree of f is*

$$\deg(f) = \sum_1^n k_i.$$

In this notation, $D \geq 0$ denotes that all $k_i \geq 0$. We can extend this so $D \geq D'$ simply means $D - D' \geq 0$.

Remark 6.4. *If the 0-superscript denotes subgroups of degree 0, then there is an exact sequence:*

$$1 \to k^* \to k(X)^* \overset{\mathrm{div}}{\to} \mathrm{Div}^0(X) \to \mathrm{Pic}^0(X) \to 0.$$

Definition 6.5 ($\mathcal{L}(D)$). *Fix $D \in Div(X)$. We can define the k-vector space*

$$\mathcal{L}(D) = \{f \in k(X)^* : \mathrm{div}(f) + D \geq 0\} \cup \{0\}, \quad \ell(D) = \dim_k \mathcal{L}(D).$$

Proposition 6.6.

$$\text{(i) } D < 0 \implies \ell(D) = 0, \quad \text{(ii) } \ell(D) < \infty, \quad \text{(iii) } D - D' = \mathrm{div}(f) \implies \mathcal{L}(D) \cong \mathcal{L}(D').$$

Proof. (i). Say $D < 0$. Any nonzero $f \in \mathcal{L}(D)$ will satisfy $\mathrm{div}(f) \geq -D > 0$. So f has no poles implying it is constant by our lemma. So $\mathcal{L}(D) = \{0\}$.

(ii). Now that we know $\ell(-p) = 0$ and divisors are finite sums, we can prove this inductively, i.e. that $\ell(D + p) \leq \ell(D) + 1$ for any point p with $\ell(D) < \infty$.

Say $D = n_p p + \sum_{q \neq p} n_q q$. For any $f \in \mathcal{L}(D + p)$, $\mathrm{ord}_p(f) + (n_p + 1) \geq 0$. If $f \notin \mathcal{L}(D)$, $\mathrm{ord}_p(f) = -n_p - 1$. If no such f exists then $\mathcal{L}(D + p) = \mathcal{L}(D)$.

If two such f, g exist, then if $(\pi) = \mathfrak{m}_p$, $\mathrm{ord}_p(f\pi^{n_p+1}) = \mathrm{ord}_p(g\pi^{n_p+1}) = 0$ so $f\pi^{n_p+1} = a + \overline{f}, g\pi^{n_p+1} = b + \overline{g}$ for $\overline{f}, \overline{g} \in \mathfrak{m}_p$. Thus, $(f - \frac{a}{b}g)\pi^{n_p+1} \in \mathfrak{m}_p$ so $\mathrm{ord}_p(f - \frac{a}{b}g) + n_p \geq 0$ so $f - \frac{a}{b}g \in \mathcal{L}(D)$, i.e. $\dim \mathcal{L}(D + p)/\mathcal{L}(D) = 1$ so $\ell(D + p) \leq \ell(D) + 1$.

(iii). Say $D = D' + \mathrm{div}(f)$. Now consider a map $\mathcal{L}(D) \to \mathcal{L}(D')$ sending $g \mapsto fg$. Clearly, $\mathrm{div}(g) + D \geq 0$ if and only if $\mathrm{div}(fg) + D' = \mathrm{div}(f) + \mathrm{div}(g) + D' = \mathrm{div}(f) + D \geq 0$ so this is well-defined. If $f = 0$, $\mathrm{div}(f) = 0$ so $D = D'$ and there is nothing to prove. Otherwise, $f \in k(X)^*$ so this map is invertible and thus an isomorphism. $\square$

Corollary 6.7. $0 \leq \ell(0) \leq 1$. *However, $\ell(0)$ contains constant functions so $\ell(0) = 1$.*

We introduce some new tools.

Definition 6.8 (Differentials). *If X is a smooth curve, the space of meromorphic differential forms Ω_X is the $k(X)$-vector space generated by symbols of the form df for $f \in k(X)$ such that, for $f, g \in k(C)$ and $a \in k$:*

$$\text{(i) } d(f + g) = df + dg, \quad \text{(i) } d(fg) = fdg + gdf, \quad \text{(i) } da = 0.$$

Proposition 6.9.

$$\dim_{k(X)} \Omega_X = 1.$$

To prove this, we need a brief lemma.

Lemma 6.10. *Every algebraic curve X over a characteristic 0 or perfect characteristic p field k is isomorphic to a planar curve, i.e. to some curve cut out by $\{f(x, y) = 0\}$ for $f \in k[x, y]$.*

Proof. (Lemma). This proof is mostly taken from a comment by Qiaochu Yuan on Math Stack-Exchange.

We use the fact, described in Theorem 5.34, that two curves are isomorphic if and only if their function fields are isomorphic.

Suppose char $k = 0$. By $\dim X = 1$, we can pick some transcendental element $x \in X$.

Suppose char $k = 0$. Then $k(X)/k$ is separable so the primitive element theorem promises the existence of $y \in k(X)$ such that $k(X) = k(x)[y]$. This is finite over $k(x)$ so y satisfies a minimal polynomial $f(x, y)$.

If k is a perfect field of characteristic p, vary x so that $L \subsetneq k(X)$ is the maximal finite separable extension of some purely transcendental extension $k(x)/k$.

Say $L \neq k(X)$. So $k(X)/L$ is purely inseparable. So we get some $f \in k(X) - L$ with $f^p \in L$ such that $y^n + \sum g_i(x)y^i \in k(x)[y]$ is the separable irreducible polynomial of f^p over $k(X)$. Since k is perfect, if $g_i \in k(x^p)$ for all i then $g_i = h_i^p$ so f is a zero of the separable polynomial $y^n + \sum h_i(x)y^i$ making f separable over $k(x)$, and thus in L - contradiction.

Hence, some $g_i(x) \notin k(x^p)$. Setting $y = f^p$, this puts x separable over $k(f)$. Since L is maximal, $f \in L$ - contradiction. So no such f exists and $k(X)/k$ is a separable and the proof from char $k = 0$ applies. □

And back to the dimension of Ω_X:

Proof. (Proposition). Observe that if $f = \frac{g}{h}$, then $fh = g$ so $hdf + fdh = dg$, i.e. $df = h^{-1}dg - fh^{-1}dh$.

We can assume by the lemma that $X = V(f)$ for $f \in k[x, y]$. Given our observation, it is obvious that $\langle x, y\rangle$ span Ω_X linearly over $k(X)$.

Of course, $0 = df = f_x dx + f_y dy$. But, by X smooth and $\dim X = 1$, both partial derivatives cannot vanish simultaneously on X. So $dx = -f_y f_x^{-1} dy$ shows that $\dim \Omega_X \leq 1$.

We show $\dim \Omega \neq 0$. That is, we produce a nonzero element. Define $S : k(X) \to k(X)$ as $g \mapsto g_x + g_y$ and extend this to $T : \Omega_X \to k(X)$ via $T(df) = S(f)$. Clearly, this satisfies all our desired relations so is well-defined. However, $T(dx) = 1 \neq 0$ (assuming without loss of generality if necessary that $x \not\equiv 0 \mod f$). □

Proposition 6.11. *For a smooth curve X, fix $p \in X$ and a uniformizer $t \in k(X)$ at p. For every $\omega \in \Omega_X$, there is a unique $f \in k(X)$ so $\omega = fdt$.*

Proof. t is transcendental over k so $k(X)/k(t)$ is finite. Similar to the proof of the above lemma, this must be separable as well.

Pick any $f \in k(X)$. Since $k(X)/k(t)$ is separable, we can get some separable polynomial $p(x, y)$ with minimal degree in y for f. Thus, $0 = dp = p_x(f, t)df + p_y(f, t)dt$ implies $df = -p_x(f, t)^{-1}p_y(f, t)^{-1}dt$. Hence, $k(X)dt = \Omega$. □

Remark 6.12. *If f is regular at p and t is a uniformizer at p, then $df = gdt$ for some unique $g \in k(X)$. Hence, df/dt is also regular.*

Say $\omega = fdt$ for uniformizers s, t at p. $ord_p(s) = ord_p(t) = 1$ so s, t are regular at p implying $dt/ds, ds/dt$ are also regular at p. Thus,

$$ord_p(\omega/dt) = ord_p(f) = ord_p(f) + ord_p(dt/ds) = ord_p((fdt)/ds) = ord_p(\omega/ds).$$

Definition 6.13 (Order). *Since $ord_p(\omega/dt)$ is independent of choice of choice of uniformizer at p by the previous remark, we define the order of ω at p as*

$$ord_p(\omega) = ord_p(\omega/dt).$$

Proposition 6.14. *Say X is a smooth curve. Fix $\omega \in \Omega_X$. For all but finitely many $p \in X$,*

$$ord_p(\omega) = 0.$$

Proof. Pick $t \in k(X) - k$ so $k(X)/k(t)$ is separable. $t \notin k$ so $k(X)/k(t)$ is finite and thus a uniformizer. Write $\omega = fdt$. Notice $f = ut^{\text{ord}_p(f)}$ and u is a unit so

$$df = (\text{ord}_p(f)ut^{\text{ord}_p(f)-1} + \frac{du}{dt}t^{\text{ord}_p(f)})dt.$$

If $g \in k(X)$, then if char $k = 0, \nmid \text{ord}_p(f)$, then $\text{ord}_p(gdf) = \text{ord}_p(g) + \text{ord}_p(f) - 1$. But, if char $k \mid \text{ord}_p(f)$, then $\text{ord}_p(gdf) \geq \text{ord}_p(g) + \text{ord}_p(f)$.

Thinking of t as a map $X \to \mathbb{P}^1$, $\deg t < \infty$ so t ramifies at finitely many points. We know from earlier work that f has finitely many zeroes and poles and t has finitely many poles yet for every other $p \in X$:

$$\text{ord}_p(\omega) = \text{ord}_p(fdt) = \text{ord}_p(fd(t - t(p))) = \text{ord}_p(f) + \text{ord}_p(t - t(p)) - 1 = 0 + 1 - 1 = 0.$$

□

Definition 6.15 (Associated divisor, holomorphic). *The divisor associated to $\omega \in \Omega_X$ is*

$$div(\omega) = \sum_{p \in X} ord_p(\omega) \cdot p \in Div(X).$$

ω is holomorphic if $div(\omega) \geq 0$.

Remark 6.16. *If $\omega_1, \omega_2 \in \Omega_X - 0$, it is clear from picking a uniformizer at some point that $\omega_1 = f\omega_2$ for $f \in k(X)$, from which $div(\omega_1) = div(f) + div(\omega_2)$.*

Definition 6.17 (Canonical divisor). *Any nonzero $\omega \in \Omega_X$ for which $K_C = div(\omega) \in Div(X)$ is a canonical divisor. Its image in $Pic(X)$ is the canonical divisor class.*

Remark 6.18.

$$\mathcal{L}(K_C) = \{f \in k(X) : \text{div}(f) + \text{div}(\omega) \geq 0\} = \{f \in k(X) : \text{div}(f\omega)\} \cong \{\eta \in \Omega_X \text{ holomorphic}\}.$$

Definition 6.19 (Genus). *We say that the genus g of a smooth curve X is $\ell(K_C) \geq 0$ for a canonical divisor K_C.*

This last definition is somewhat circular, but the point is that we get the following theorem:

Theorem 6.20 (Cheap Riemann-Roch). *Say X is a smooth curve with a canonical divisor K_C. There is an integer $g \geq 0$ (the genus) such that for all $D \in Div(X)$:*

$$\ell(D) - \ell(K_C - D) = \deg D - g + 1.$$

Corollary 6.21. $\deg K_C = 2g - 2$. *If $\deg D > 2g - 2$, then $\ell(D) = \deg D - g + 1$.*

Definition 6.22 (Elliptic curve). *An elliptic curve is a smooth curve X of genus one with a fixed base-point O.*

Proposition 6.23. *Say (X,O) is an elliptic curve over k. There is a map $\phi : (X,O) \to (\mathbb{P}^2, [0:1:0])$ given by $[x:y:1]$ for $x, y \in k(X)$ such that ϕ is an isomorphism onto a smooth curve cut out by a 'Weierstraß equation'*

$$y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6, \quad a_i \in k.$$

Proof. From the above corollary to Riemann-Roch, $\ell(nO) = \deg(nO) = n$ for $n > 2g-2 = 0$. Thus, we can pick $1, x, y \in k(X)$ so $(1,x)$ span $\ell(2O)$ and $(1,x,y)$ span $\ell(3O)$.

Necessarily, x, y will have poles at O of order 2 and 3 respectively. Thus, the seven $1, x, y, xy, x^2, y^2, x^3$ will all have poles of order at most 6 at O, putting them in $\mathcal{L}(6O)$. However, $\ell(6O) = 6$ (where this is dimension over k) so there is a nontrivial relation

$$A_1 1 + A_2x + A_3y + A_4xy + A_5x^2 + A_6y^2 + A_7x^3 = 0, \quad A_i \in k.$$

Changing into $x = -A_6A_7x', y = A_6A_7^2y'$ gives:

$$A_1 - A_2A_6A_7x' + A_3A_6A_7^2y' - A_4A_6^2A_7^3x'y' + A_6^2A_7^2(x')^2 + A_6^3A_7^4(y')^2 - A_6^3A_7^4(x')^3 = 0.$$

Neither A_6, A_7 can be 0 since $1, x, y, xy, x^2, y^2, x^3$ have poles of order (respectively) $0, 2, 3, 5, 4, 6, 6$ so if either A_6 or A_7 was 0, all the other A_i would vanish - contradiction. So dividing by $A_6^3A_7^4$ returns the desired Weierstraß equation.

Notice $\phi = [x' : y' : 1]$ has image in the curve defined by the null-set of this equation. It is clearly a morphism so its image is a projective variety in $\mathbb{P}^2$ and thus a curve. Since ϕ is non-constant, it is surjective. Observe $[k(X) : k(x')] = 2$ and $[k(X) : k(y')] = 3$ so $[k(X) : k(x,y)] = 1$ so by the corollary to the equivalence of categories in Theorem 5.34, ϕ is an isomorphism onto its image, making its image also a smooth curve. $\square$

Remark 6.24. *As long as char $k \neq 2$ (in which case it is either prime and odd or 0), we can set $y' = y + \frac{a_1}{2}x + \frac{a_3}{2}$ and $x' = x$. Thus:*

$$y^2 + a_1xy + a_3y = [(y')^2 - a_1x'y' - a_3y'] + [a_1x'y'] + [a_3y'] + F(x') = (y')^2 + F(x').$$

This allows us to reduce the Weierstraß equation to

$$y^2 = x^3 + a_2x^2 + a_4x + a_6.$$

But changing x to $x' = x + \frac{a_2}{3}$ (if char $k \neq 3$ also) similarly reduces to

$$y^2 = x^3 + ax + b.$$

Corollary 6.25. *It follows that any elliptic curve over $k = \mathbb{C}$ is a compact Riemann surface, and by its function field, is indeed an elliptic Riemann surface!*

Remark 6.26. *The converse holds but is more difficult and requires Dirichlet's principle or other hard analysis (sometimes used in the proof of Riemann-Roch) so we omit it. [Sil86]*

§7. Vector Bundles

In this section, we explore vector bundles.

Definition 7.1 (Fiber bundle). *A fiber bundle is a tuple (E, X, F, p) where p is a continuous surjection from the 'total space' E to the 'base space' X such that for all $x \in X$, there is a neighborhood U_x of x and homeomorphism $\phi_x : p^{-1}U_x \to U_x \times F$ for which $p|_{p^{-1}U_x} = \mathrm{proj}_{U_x} \circ \phi_x$.*

Remark 7.2. *It follows that the fiber over x is $p^{-1}(x) = \phi_x^{-1} \circ \mathrm{proj}_{U_x}^{-1}(x) = \phi_x^{-1}(\{x\} \times F) \cong F$.*

p is open since $\{\phi_x\}_x$ are homeomorphisms and $\{\mathrm{proj}_{U_x}\}_x$ are open. So p is a quotient map.

Definition 7.3 (Structure group). *Observe $\phi_x \circ \phi_y^{-1}(p,v) = (p, g_{xy}(p)(v))$ for a continuous map $g_{xy} : U_x \cap U_y \to G$ to a topological group G which acts on k^r by homeomorphisms.*

Definition 7.4 (Vector bundle). *A vector bundle is a fiber bundle (E, X, F, p) whenever F is a finite-dimensional k-vector space and the homeomorphisms $\phi_x|_{p^{-1}(x)} : p^{-1}(x) \to F$ are k-linear.*

Remark 7.5. *Notice each fiber $p^{-1}(x)$ in a vector bundle gets the structure of a k-vector space.*

Definition 7.6 (Algebraic vector bundle). *An algebraic vector bundle (E, X, F, p) is a vector bundle for projective varieties E, X and regular maps $p, \{\phi_x\}_x$. (Also, k^r is replaced by $\mathbb{A}^r$).*

Remark 7.7. *Bundle morphisms are fiber-wise and regular.*

Remark 7.8. *Given that $Gl_r(k)$ is itself a variety cut out by the nonzero determinant condition, the $g_{xy} : U_x \cap U_y \to Gl_r(k)$ will be a morphism of varieties as well.*

For later use, note the following definition too.

Definition 7.9 (Indecomposable). *A bundle E is indecomposable if $E \cong F \oplus G$ implies either F or G is isomorphic to E and the other one is 0.*

§8. Sheaves

Acknowledgment. *This section vaguely follows [Har77].*

Now we compare with the sheaf-theoretic picture.

Definition 8.1 (Presheaf). *A presheaf $\mathcal{F}$ of rings on a topological space X is a functor from the open sets of X, $Op(X)^{op}$, to Ring. In particular, $V \hookrightarrow U$ implies the existence of a 'restriction' ring homomorphism $res^U_V : \mathcal{F}(U) \to \mathcal{F}(V)$ such that:*

$$res^U_U = \mathrm{id}_{\mathcal{F}(U)} \quad \text{and} \quad W \subseteq V \subseteq U \implies res^U_W = res^V_W \circ res^U_V.$$

Definition 8.2 (Section). *Any $s \in \mathcal{F}(U)$ is a section of $\mathcal{F}$ over U; any $s \in \mathcal{F}(X)$ is a global section.*

Remark 8.3. *The space of sections $\mathcal{F}(U)$ is also denoted $\Gamma(U, \mathcal{F})$.*

Remark 8.4. *We often use $s|_V$ instead of $res^U_V(s)$ for $s \in \mathcal{F}(U)$.*

Definition 8.5 (Sheaf). *A presheaf $\mathcal{F}$ is a sheaf if it satisfies the following:*

(i). *Say $U = \cup_\alpha U_\alpha$ for U_α open. Whenever $s, t \in \mathcal{F}(U)$ and $s|_{U_\alpha} = t|_{U_\alpha}$ for all α, then $s = t$.*

(ii). *Say $U = \cup_\alpha U_\alpha$ for U_α open. Given $\{s_\alpha \in \mathcal{F}(U_\alpha)\}_\alpha$ such that $s_\alpha|_{U_\alpha \cap U_\beta} = s_\beta|_{U_\alpha \cap U_\beta}$ for all α, β, then there is, by* (i) *a unique, $s \in \mathcal{F}(U)$ so $s|_{U_\alpha} = s_\alpha$.*

Example 8.6. *These extra conditions may seem obvious but do not always hold.*

- (i) *may fail while* (ii) *does not. Consider a presheaf $\mathcal{F}$ on some space X with $\mathcal{F}(U \subsetneq X) = 0$, $\mathcal{F}(X) = \mathbb{Z}/2\mathbb{Z}$, $res^{U \neq X}_V = 0$, and $res^X_{V \neq X} = 0$. If $X = \cup_\alpha U_\alpha$ for $U_\alpha \subsetneq X$, then of course $s = 0, t = 1 \in \mathcal{F}(X)$ will agree on the restriction to U_α, but $s \neq t$.*
- (ii) *may fail while* (i) *does not. Consider a presheaf $\mathcal{F}$ of bounded holomorphic functions on $\mathbb{C}$ with the obvious restriction morphisms. Say $\mathbb{C} = \cup_{n \in \mathbb{N}} B_n(0)$ and pick a family $\{s_n = \mathrm{id}_{B_n(0)} \in \mathcal{F}(B_n(0))\}_{n \in \mathbb{N}}$. For any $n \leq m$, $s_n|_{B_n(0)} = s_m|_{B_n(0)}$, but all bounded holomorphic functions on $\mathbb{C}$ are constant so will not restrict to any of the $s_{n>0}$.*

Definition 8.7 (Stalk, germ). *The stalk of a sheaf $\mathcal{F}$ at $x \in X$ is the direct limit*

$$\mathcal{F}_x = \varinjlim_{x \in U} \mathcal{F}(U) = \left(\coprod_{x \in U} \mathcal{F}(U) \right) / \sim, \qquad \text{where}$$

$$s \in \mathcal{F}(U) \sim t \in \mathcal{F}(V) \text{ if and only if } s|_W = t|_W \text{ for } W \subseteq U \cap V.$$

Given U containing x, the usual map $\mathcal{F}(U) \to \mathcal{F}_x$ sends any $s \in \mathcal{F}(U)$ in $\mathcal{F}_x$ to its germ at x.

Definition 8.8 (Sheaf morphism). *A morphism of sheaves $f : \mathcal{F} \to \mathcal{G}$ over a space X is a collection of morphisms $f(U) : \mathcal{F}(U) \to \mathcal{G}(U)$, e.g. ring homomorphisms, such that:*

$$\begin{tikzcd}
\mathcal{F}(U) \arrow[r, "f(U)"] \arrow[d, "res^U_V"'] \arrow[dr, phantom, "\circlearrowleft"] & \mathcal{G}(U) \arrow[d, "res^U_V"] \\
\mathcal{F}(V) \arrow[r, "f(V)"'] & \mathcal{G}(V)
\end{tikzcd}$$

Remark 8.9. *A morphism of sheaves $f : \mathcal{F} \to \mathcal{G}$ induces a morphism on stalks $f_x : \mathcal{F}_x \to \mathcal{G}_x$.*

We generalize the Zariski topology discussed earlier as follows:

Definition 8.10 (Spectrum). *Given a ring R, its spectrum $X = Spec(R) = \{\mathfrak{p} \lhd R \text{ prime}\}$ is a topological space under a basis of open sets of the form $X_I = \{\mathfrak{p} : I \not\subseteq \mathfrak{p}\}$ for all ideals $I \lhd R$.*

Proof. We check this is a topological space. $X = X_1$ and $\varnothing = X_0$. Since we are considering prime ideals, $X_I \cap X_J = \{\mathfrak{p} : I, J \not\subseteq \mathfrak{p}\} = X_{IJ}$. Given $\{I_\alpha\}_\alpha$ and $I = \sum_\alpha I_\alpha$, $\cup_\alpha X_{I_\alpha} = \{\mathfrak{p} : I_\alpha \not\subseteq \mathfrak{p} \text{ for some } \alpha\} = X_I$ again by considering prime ideals. $\square$

Remark 8.11. *$\{X_f\}_{f \in R}$ clearly forms a basis for X. It suffices to define sheaves on this basis.*

Definition 8.12 (Structure sheaf). *The structure sheaf $\mathcal{O}_X$ assigns $\mathcal{O}_X(X_f) = R_f = R[1/f]$.*

The morphisms are trickier. If $X_f \subseteq X_g$, then $\{\mathfrak{p} : (g) \subseteq \mathfrak{p}\} \subseteq \{\mathfrak{p} : (f) \subseteq \mathfrak{p}\}$ so the intersection of all primes containing (g) will also contain (f), i.e. $f \in \sqrt{(g)}$ so $f^k = g$ for some k. Hence, we get a restriction homomorphism $R[1/g] \to R[1/f]$ via $\frac{r}{g^n} \mapsto \frac{r}{f^{nk}}$.

Proposition 8.13. *The stalks in X are*

$$\mathcal{O}_\mathfrak{p} = \varinjlim_{\mathfrak{p} \in X_f} \mathcal{O}_X(X_f) = \varinjlim_{f \notin \mathfrak{p}} R_f = R_\mathfrak{p}.$$

Proof. Everything is by definition except the last equality. From our description of the restriction homomorphisms, it is obvious that the projection $\phi : \coprod_{f \notin \mathfrak{p}} R_f \twoheadrightarrow R_\mathfrak{p}$ descends to a ring homomorphism $\phi : \varinjlim_{f \notin \mathfrak{p}} R_f \twoheadrightarrow R_\mathfrak{p}$. If $\phi(\frac{a}{f^n}) = 0$ for some $f \notin \mathfrak{p}$, then $\frac{a}{f^n} \cdot \frac{g}{g} = \frac{0}{f^n g}$ so there is some $g \notin \mathfrak{p}$ such that $ag = 0$. However, $fg \notin \mathfrak{p}$ so $0 = \frac{ag^n}{f^n g^n} \in X_{fg} = X_f \cap X_g$, meaning $\frac{a}{f^n}$ was eventually 0 in the limit already. Hence ϕ is injective so an isomorphism. $\square$

Thus, the structure sheaf fits a slightly more general definition.

Definition 8.14 ((Locally) Ringed). *A space X is ringed if it is equipped with a sheaf of rings $\mathcal{F}$. It is locally ringed if each stalk $\mathcal{F}_x$ is a local ring.*

Remark 8.15. *We can apply this theory to the case of an algebraic smooth curve X.*

Recall $k[X] = k[x_0, \dots, x_n]/I(X)$ is one-dimensional, i.e. any nonzero prime (i.e. containing $I(X)$) must be maximal, and thus corresponds to a point on the variety. So we can write

$$Spec\ k[X] = \{0\} \cup \{\mathfrak{m}_p\}_{p \in X} = \{0\} \cup \{\{f \in k[X] : f(p) = 0\}\}_{p \in X}.$$

Because $k[X]$ is Noetherian, any ideal $I = (f_1, \dots, f_j)$ for $f_i \not\equiv 0 \mod I(X)$. Since k is algebraically closed, we can find the finite number of collective roots $p_1, \dots, p_\ell$ of the f_i. So any open set $X_I = \cup_1^j X_{f_i}$ really is $X - \{\mathfrak{m}_{p_r}\}_1^\ell$. It should be clear $Spec\ k[X]$ is homeomorphic to X.

We can treat the structure sheaf $\mathcal{O}_{Spec\ k[X]}$ as the structure sheaf $\mathcal{O}_X$ on X where, by the above proposition, each stalk is exactly the local ring $k[X]_p$ for p a point on X or $k[X]_0 = k(X)$.

Definition 8.16 (Sheaf of $\mathcal{O}_X$-modules). *A sheaf of $\mathcal{O}_X$-modules on a ringed space $(X, \mathcal{O}_X)$ is a sheaf $\mathcal{F}$ such that each $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$-module and for all $f, g \in \mathcal{F}(U)$, $s \in \mathcal{O}_X(U)$:*

$$(f+g)|_V = f|_V + g|_V \quad \text{and} \quad (sf)|_V = s|_V f|_V \quad \text{where} \quad V \subseteq U.$$

Definition 8.17 (Morphism of $\mathcal{O}_X$-modules). *$f : \mathcal{F} \to \mathcal{G}$ is a morphism of $\mathcal{O}_X$-modules for $\mathcal{O}_X$-modules $\mathcal{F}, \mathcal{G}$ as long as every $f(U)$ is an $\mathcal{O}_X(U)$-module homomorphism.*

There is some subtlety about 'sheafifications' that we ignore but, essentially all of the usual module operations hold for sheaves in the obvious way:

Remark 8.18 ($\mathcal{O}_X$-module operations). *The following are all $\mathcal{O}_X$-modules:*

$$\mathcal{F}\otimes_{\mathcal{O}_X}\mathcal{G}, \quad \mathcal{F}\oplus\mathcal{G}, \quad \mathcal{F}\times\mathcal{G}, \quad \underset{\mathcal{G}<\mathcal{F}}{\mathcal{F}/\mathcal{G}}, \quad \varinjlim \mathcal{F}_\alpha, \quad \varprojlim F_\alpha, \quad \mathit{Hom}_{\mathcal{O}_X}(\mathcal{F},\mathcal{G}), \quad \mathcal{F}^\vee = \mathit{Hom}_{\mathcal{O}_X}(\mathcal{F},\mathcal{O}).$$

Definition 8.19 ((Locally) free sheaf). *An $\mathcal{O}_X$-module $\mathcal{F}$ is a free sheaf if it is isomorphic as an $\mathcal{O}_X$-module to $\oplus^r\mathcal{O}_X$. $\mathcal{F}$ is a locally free sheaf if every $\mathcal{F}|_{U_\alpha}$ is a free sheaf for an open cover $\{U_\alpha\}_\alpha$ of X, where $\mathcal{F}|_U$ is $\mathcal{F}$ restricted to the open subsets of some open set U in X.*

Definition 8.20 (Rank). *The rank of a free sheaf $\mathcal{F} \cong \mathcal{O}_X^r$ is r (r can be ∞).*

Remark 8.21. *The rank of free sheaves is constant over connected components.*

Remark 8.22. *If $\mathcal{F}$ is locally free of rank $r < \infty$, then $\mathcal{F}^\vee$ is locally free of rank r.*

Definition 8.23 (Invertible sheaf). *A locally free sheaf of rank 1 is an invertible sheaf.*

Proposition 8.24. *If $\mathcal{F}$ is an invertible sheaf of $\mathcal{O}_X$-modules, then*

$$\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{F}^\vee \cong \mathcal{O}_X.$$

Proof. We can cover $X = \cup U_\alpha$ so that $\mathcal{F}|_{U_\alpha} \cong \mathcal{O}_X|_{U_\alpha}$. Therefore,

$$(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{F}^\vee)|_{U_\alpha} = \mathcal{F}|_{U_\alpha} \otimes_{\mathcal{O}_X|_{U_\alpha}} \mathrm{Hom}(\mathcal{F}|_{U_\alpha}, \mathcal{O}_X|_{U_\alpha}) \cong \mathcal{O}_X|_{U_\alpha} \otimes_{\mathcal{O}_X|_{U_\alpha}} \mathrm{Hom}(\mathcal{O}_X|_{U_\alpha}, \mathcal{O}_X|_{U_\alpha}).$$

It suffices to show $\mathcal{O}_X \otimes_{\mathcal{O}_X} \mathcal{O}_X^\vee \cong \mathcal{O}_X$ as $\mathcal{O}_X$-modules. The map $f(U) : s \otimes t \mapsto t(s)$ defines a sheaf morphism $f : \mathcal{F} \otimes \mathcal{F}^\vee \to \mathcal{O}_X$ since

$$f(U)(s \otimes t)|_V = (t(s))|_V = t|_V(s|_V) = f(V)(s|_V \otimes t|_V).$$

$f(U)$ is an $\mathcal{O}_X(U)$-homomorphism since t is $\mathcal{O}_X(U)$-linear. Hence f is a morphism of $\mathcal{O}_X$-modules. It is obviously an isomorphism.

We need only show that $(\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{F}^\vee)|_{U_\alpha} \cong \mathcal{O}_X|_{U_\alpha}$ implies $\mathcal{F} \cong \mathcal{O}_X$, but this is obvious by the definition of sheaves (gluing). □

Lemma 8.25. *If $\mathcal{F}, \mathcal{G}, \mathcal{H}$ are all locally free $\mathcal{O}_X$-modules of finite rank, then*

$$(\mathcal{F} \otimes \mathcal{G}) \otimes \mathcal{H} \cong \mathcal{F} \otimes (\mathcal{G} \otimes \mathcal{H}), \quad \mathcal{F} \otimes \mathcal{G} \cong \mathcal{G} \otimes \mathcal{F}.$$

Proof. We omit the definition of 'sheafification,' but morally this follows from the associativity and commutativity of tensor products for $\mathcal{O}_X(U)$-modules. □

We note an additional trivial lemma.

Lemma 8.26. *If $\mathcal{F}, \mathcal{G}$ are locally free of rank one, then $\mathcal{F} \otimes \mathcal{G}$ is locally free of rank one.*

The last proposition and two lemmas let us define the following group.

Definition 8.27 (Picard group). *Up to isomorphism, the invertible $\mathcal{O}_X$-sheaves on X form an abelian group $\mathrm{Pic}\, X$ under $\otimes_{\mathcal{O}_X}$ called the Picard group of $(X, \mathcal{O}_X)$.*

We can finally make the equivalence between bundles and locally free sheaves explicit.

Fix an algebraic vector bundle (E, X, F, p).

Definition 8.28 ($\mathcal{O}_X(E)$). *Pick a basic open set $X_f \subseteq X$. Define the space of regular sections*

$$\mathcal{O}_X(E)(X_f) = \{s : X_f \to E : s \text{ is regular and } p \circ s = \mathrm{id}_{X_f}\}.$$

The restriction maps $\mathcal{O}_X(E)(X_g) \to \mathcal{O}_X(E)(X_f)$ induced by $X_f \hookrightarrow X_g$ are the usual function restriction (and of course well-defined with respect to poles).

Proposition 8.29. *$\mathcal{O}_X(E)$ is a locally free rank $\mathrm{rk}_k E$ sheaf of $\mathcal{O}_X$-modules.*

Proof. If $a \in \mathcal{O}_X(X_f) = k[X]_f$, then a only has poles at the roots of f and X_f excludes these roots. Thus, $\mathcal{O}_X(E)(X_f)$ forms an $\mathcal{O}_X(X_f)$-module under the action

$$(s+t)(p) = s(p) + t(p) \quad \text{and} \quad (as)(p) = a(p)s(p) \quad \text{for } s, t \in \mathcal{O}_X(E)(X_f).$$

$\mathcal{O}_X(E)$ obviously satisfies locality. Moreover, if $U = \cup U_\alpha$ and $s_\alpha \in \mathcal{O}_X(E)(U_\alpha)$ agree on overlaps, then we can define a $s : U \to E$ as $s(p) = s_\alpha(p)$ for $p \in U_\alpha$. s is continuous by gluing lemma and thus regular since we assumed that it is locally rational. $p \circ s = \mathrm{id}_U$ by definition of s so $\mathcal{O}_X(E)$ is indeed sheaf.

Because E is a vector bundle over a compact space, there are finitely many local trivializations $(U_i)_1^n$ for $E \twoheadrightarrow X$ such that $p^{-1}U_i \cong U_i \times \mathbb{A}^r$. Fixing this trivialization, there are r canon sections $U_i \to U_i \times \mathbb{A}^r \cong p^{-1}U_i \subseteq E$ so $\mathcal{O}_X(E)(U_i) \cong \mathcal{O}_X(U_i)^r$. X is connected so $\mathcal{O}_X(E)$ is a locally free sheaf of rank $\mathrm{rk}_k E$. $\square$

We show the other direction also works. Suppose $\mathcal{F}$ is a rank r locally free $\mathcal{O}_X$-sheaf on X.

Definition 8.30 ($E(\mathcal{F})$). *There are finitely many trivializations $(U_i)_1^n$ covering X such that $\mathcal{F}|_{U_i} \cong (\mathcal{O}_X|_{U_i})^r$. Notice the composition of sheaf isomorphisms*

$$(\mathcal{O}_X|_{U_i \cap U_j})^r = (\mathcal{O}_X|_{U_i})^r|_{U_i \cap U_j} \xrightarrow{g_i^{-1}} \mathcal{F}_{U_i}|_{U_i \cap U_j} = \mathcal{F}_{U_j}|_{U_i \cap U_j} \xrightarrow{g_j} (\mathcal{O}_X|_{U_j})^r|_{U_i \cap U_j} = (\mathcal{O}_X|_{U_i \cap U_j})^r.$$

So $g_{ij} = g_j \circ g_i^{-1}$ is an automorphism of $(\mathcal{O}_X|_{U_i \cap U_j})^r$ – an invertible $r \times r$-matrix. Define

$$E(\mathcal{F}) = \coprod_1^n U_i \times \mathbb{A}^r / \left((x, v) \sim (x, g_{ij}(v) \right).$$

Proposition 8.31. *$E(\mathcal{F}) \twoheadrightarrow X$ is a rank r vector bundle over X under the obviously continuous projection to X. $E(\mathcal{F})$ is a projective variety since it is covered by affines per construction.*

Remark 8.32. *We could have defined $E(\mathcal{F})$ differently. For all $p \in X$, $\mathfrak{m}_p$ is the maximum ideal of local ring $k[X]_p$. The stalk $\mathcal{F}_p$ is an $(\mathcal{O}_X)_p = k[X]_p$-module. Thus, $\mathcal{F}_p/\mathfrak{m}_p\mathcal{F}_p$ is a $k[X]_p/\mathfrak{m}_p$ vector space. Define*

$$E(\mathcal{F}) = \coprod_{p \in X} \mathcal{F}_p/\mathfrak{m}_p\mathcal{F}_p \twoheadrightarrow X, \quad s \in \mathcal{F}_p/\mathfrak{m}_p \mapsto p.$$

The remaining work comes down to making $E(\mathcal{F})$ a projective variety.

We summarize these results as follows.

Theorem 8.33. *The categories of rank r vector bundles is equivalent to the category of rank r locally free sheaves over a projective variety.*

Proof. $E \cong E(\mathcal{O}_X(E))$ and $\mathcal{F} \cong \mathcal{O}_X(E(\mathcal{F}))$ are canonical, natural isomorphisms. $\square$

§9. Line Bundles

In this section, we use the categorical equivalence to explore rank one vector bundles.

Definition 9.1 (Line bundles). *Line bundles are the bundles associated to invertible sheaves.*

Definition 9.2 ($\mathcal{O}_X(D)$). *Fix a divisor D on X. For basic open sets X_f, define the presheaf*

$$\mathcal{O}_X(D)(X_f) = \{g \in \mathcal{O}_X(X_f) = k[X]_f : \mathrm{div}(g) + D \geq 0\}.$$

Using the obvious restriction maps, this extends to a sheaf $\mathcal{O}_X(D)$ for the same reasons $\mathcal{O}_X$ did.

Lemma 9.3. *$\mathcal{O}_X(D)$ is a line bundle.*

Proof. $\mathcal{O}_X(D)$ is equal to $\mathcal{O}_X$ on any open set X_f which does not meet D. And it is isomorphic on the sets which do. These isomorphisms are compatible with the equalities and together the sets cover X so we are done. $\square$

Let us give an inverse to $D \mapsto \mathcal{O}(D)$.

Remark 9.4. *Fix a line bundle L on X. Pick finitely many local trivializations $(U_i)_1^n$ of L. $L(U_i) \cong \mathcal{O}_X(U_i)$ is an $\mathcal{O}_X$-isomorphism so some $f_i \in \Gamma(U_i, L)$ maps to $1 \in \Gamma(U_i, \mathcal{O}_X)$. Set*

$$D_L = \sum_{p \in X} \min_{1 \leq i \leq n} ord_p(f_i)p.$$

L is clearly isomorphic to $\mathcal{O}_X(D_L)$.

Lemma 9.5. *$\mathcal{O}_X(D) \otimes \mathcal{O}_X(D') \cong \mathcal{O}_X(D+D')$ for any divisors D under $f \otimes g \mapsto fg$.*

Proof. Obvious. $\square$

Corollary 9.6. *In section 8, we showed that invertible sheaves / line bundles form an abelian group under $\otimes_{\mathcal{O}_X}$. Since $\mathcal{O}_X(0) = \mathcal{O}_X$, the lemma actually implies that*

$$\mathcal{O}_X(-D) \cong \mathcal{O}_X(D)^\vee.$$

Proposition 9.7. *$\mathcal{O}_X(D) \cong \mathcal{O}_X$ if and only if $D = div(f)$ for some $f \in k(X)$.*

Proof. $\implies$) : Suppose $\mathcal{O}_X(D) \cong \mathcal{O}_X$. The global section $1 \in \mathcal{O}_X(X)$ must come from a nonzero global section $f \in \mathcal{O}_X(D)(X)$, i.e. $\operatorname{div}(f) + D \geq 0$. Notice $1/f \in \mathcal{O}_X(-D)(X)$ so $\operatorname{div}(1/f) - D \geq 0$. But $\operatorname{div}(1/f) = -\operatorname{div}(f)$ implies $0 \geq \operatorname{div}(f) + D \geq 0$ so $D = \operatorname{div}(1/f)$ is principal.

$\impliedby$) : Suppose $D = \operatorname{div}(f)$ for some $f \in k(X)^*$. Then $\mathcal{O}_X(U) \to \mathcal{O}_X(D)(U)$ multiplying sections by f (with the inverse multiplying by $1/f$) is an isomorphism. $\square$

Theorem 9.8. *Using the definition of the Picard group from section 8,*

$$\text{Div } X/\text{principal divisors} \cong \text{Pic } X \quad \textit{is a group isomorphism.}$$

Proof. $D \mapsto \mathcal{O}_X(D)$ is a group homomorphism per section 8 and the above lemma. This map is surjective since any representative L of a class of line bundle in Pic X is isomorphic to some $\mathcal{O}(D_L)$. Lastly, the kernel is precisely the principal divisors PDiv X. That is, if $\mathcal{O}_X(D) \cong \mathcal{O}_X(D')$, then the lemma also implies

$$\mathcal{O}_X(D - D') \cong \mathcal{O}_X(D) \otimes \mathcal{O}_X(-D') \cong \mathcal{O}_X(0) = \mathcal{O}_X.$$

Yet the previous proposition says this holds if and only if $D - D' = \operatorname{div}(f)$ for some $f \in k(X)$. $\square$

Because principal divisors have degree 0, this next word is well-defined.

Definition 9.9 (Degree). *The <u>degree</u> of a line bundle L is $\deg D_L$.*

If $\mathcal{F}$ is a locally free sheaf of rank r, then its <u>determinant bundle</u> $\det \mathcal{F} = \bigwedge^r \mathcal{F}$ is of rank $\binom{r}{r} = 1$, making it into a line bundle. We say $\deg \mathcal{F} = \deg D_{\bigwedge^r \mathcal{F}}$.

This theorem is immediately applicable.

Theorem 9.10. Pic $\mathbb{P}^1 \cong \mathbb{Z}$ *as groups.*

Proof. Let $X = \mathbb{P}^1$ where $k[X]$ is the collection of homogeneous polynomials in x, y.

Per theorem 9.8, it is enough to show the isomorphism of $\mathbb{Z}$ with divisors up to a principal divisor. First recall that

$$k(X) = \left\{ \frac{f}{g} : f, g \in k[x, y] \text{ homogeneous and of same degree}, g \neq 0 \right\}.$$

Fix any distinct $p, q \in \mathbb{P}^1$. The coordinate chart $x \neq 0$ will include at least one of p, q. If it has both, multiplying any $f \in k(X)$ by $\frac{y/x - \tilde{p}}{y/x - \tilde{q}}$ will give divisor $\operatorname{div}(f) + p - q$. If the chart only has one, pick a third point $r \in X_{xy}$, and use the same fraction to swap p for r, which reduces to the case of the previous sentence. Seeing that divisors of every degree are attainable, this shows all divisors of the same degree differ by a principal divisor. □

§10. Riemann-Roch

Acknowledgment. *This section uses the definition of sheaf cohomology from [WG08].*

We briefly define the cohomology of a sheaf $\mathcal{F}$ of abelian groups over a topological space X and present a more generalized version of Riemann-Roch and Serre duality than as presented in section 6.

Remark 10.1. *Fix an open cover $\mathcal{U}$ of X.*

Definition 10.2 (q-simplex). *An ordered-tuple $\sigma = (U_0, \dots, U_q)$ for $U_i \in \mathcal{U}$ is a q-simplex as long as its support $|\sigma| = \cap_0^q U_i$ is nonempty.*

Definition 10.3 (q-cochain). *A q-cochain of $\mathcal{F}$ is a map f sending any q-simplex σ to $f(\sigma) \in \mathcal{F}(|\sigma|)$. The abelian group (under pointwise addition) of q-cochains is denoted $C^q(\mathcal{U}, \mathcal{F})$.*

Definition 10.4 (Coboundary). *If $f \in C^q(\mathcal{U}, \mathcal{F})$ and $\sigma = (U_0, \dots, U_{q+1})$, the coboundary operator $\delta : C^q(\mathcal{U}, \mathcal{F}) \to C^{q+1}(\mathcal{U}, \mathcal{F})$ acts by*

$$\delta f(\sigma) = \sum_0^{q+1} (-1)^i f((U_0, \dots, \hat{U}_i, \dots, U_{q+1}))|_{\cap_{j \neq i} U_j}.$$

Remark 10.5. *δ is clearly a group homomorphism. And if $\sigma = (U_0, \dots, U_{q+2})$, then $\delta^2 f(\sigma)$ is*

$$= \sum_0^{q+1} (-1)^i \left[\sum_{j=0}^{i-1} (-1)^j f(U_0, \dots, \hat{U}_j, \dots, \hat{U}_i, \dots, U_{q+2})|_{U_{\neq i,j}} + \sum_{j=i+1}^{q+2} (-1)^{j-1} f(U_0, \dots, \hat{U}_i, \dots, \hat{U}_j, \dots, U_{q+2})|_{U_{\neq i,j}} \right]$$

$$= \sum_{i,j} \left[(-1)^{i+j} f(U_0, \dots, \hat{U}_j, \dots, \hat{U}_i, \dots U_{q+2}) + (-1)^{i+j-1} f(U_0, \dots, \hat{U}_j, \dots, \hat{U}_i, \dots U_{q+2}) \right] = 0.$$

This induces a cochain complex:

$$C^\bullet(\mathcal{U}, \mathcal{F}) = C^0(\mathcal{U}, \mathcal{F}) \xrightarrow{\delta} C^1(\mathcal{U}, \mathcal{F}) \xrightarrow{\delta} C^2(\mathcal{U}, \mathcal{F}) \xrightarrow{\delta} C^3(\mathcal{U}, \mathcal{F}) \xrightarrow{\delta} \cdots$$

Hence, we can define cohomology

$$\check{H}^q(\mathcal{U}, \mathcal{F}) = \check{H}^q(C^\bullet(\mathcal{U}, \mathcal{F})) = Z^q(\mathcal{U}, \mathcal{F}) / B^q(\mathcal{U}, \mathcal{F}).$$

Given a refinement $\mathcal{V}$ of $\mathcal{U}$, the restriction maps of $\mathcal{F}$ induce group maps $C^q(\mathcal{U}, \mathcal{F}) \to C^q(\mathcal{V}, \mathcal{F})$.

Definition 10.6 (Čech cohomology). *The Čech cohomology of X with coefficients in $\mathcal{F}$ is*

$$\check{H}^q(X, \mathcal{F}) \cong \varinjlim_{\mathcal{U}} \check{H}^q(\mathcal{U}, \mathcal{F}).$$

Remark 10.7. *On varieties, Čech cohomology is isomorphic to sheaf cohomology.*

Finally, we can state Riemann-Roch and Serre duality in full.

Theorem 10.8 (Riemann-Roch). *A vector bundle E on a smooth genus g curve X satisfies:*

$$\dim_k H^0(X, E) - \dim_k H^0(X, \Omega_X \otimes E^\vee) = \deg E + (rkE)(1 - g).$$

Theorem 10.9 (Serre duality). *There is a perfect pairing*

$$H^0(X, \Omega_X \otimes L^\vee) \times H^1(X, L) \to k.$$

So $\dim_k H^1(X, L) = \dim_k H^0(X, \Omega_X \otimes L^\vee)$.

Corollary 10.10. *For elliptic curves, Riemann-Roch is simply*

$$\dim_k H^0(X, E) - \dim_k H^1(X, E) = \deg E.$$

Remark 10.11. *Observe that* $\check{H}^0(X, \mathcal{F})$ *is by definition every possible assignment from* X *to* $\mathcal{F}(X)$*, i.e.* $\check{H}^0(X, \mathcal{F}) = C^0(X, \mathcal{F}) = \mathcal{F}(X) = \Gamma(X, \mathcal{F})$ *is the ring of global sections of* $\mathcal{F}$*.*

The spaces of rational functions $\mathcal{L}(D)$ *defined in section 6 are exactly* $H^0(X, \mathcal{O}_X(D))$*.*

Now, we can do a one-higher genus version of theorem 9.10.

Theorem 10.12. *The degree 0 subgroup of Pic* X *is called the* <u>*Picard variety Pic*$^0 X$</u>*.*

For an elliptic curve (X, O)*, Pic*$^0 X \cong X$ *is a bijection. (And if* X *comes from a Weierstraß equation, then the induced group law on* X *matches that of Pic*^{0} X*.)*

Proof. We work with the definition of Picard group from theorem 9.8.

First, observe that since $\deg \Omega_X = 2g - 2 = 0$,

$$\dim H^1(\mathcal{O}_X(D > 0)) = \dim H^0(\Omega_X \otimes \mathcal{O}_X(D > 0)^\vee) = \dim H^0(\Omega_X \otimes \mathcal{O}_X(-D < 0)) = 0.$$

So by Riemann-Roch, $\dim H^0(\mathcal{O}_X(D > 0)) = \deg D$.

Consider the map $X \to \operatorname{Pic}^0 X$ via $p \mapsto p - O$. If $p - O = q - O + \operatorname{div}(f)$ for some $f \in k(X)$, then $f \in \mathcal{O}_X(1 \cdot q)$. By the above comment, this space if one-dimensional so $f \neq 0$ will span. But obviously not all functions in $\mathcal{O}_X(q)$ must vanish at p so $f = 0$ and $p = q$, showing injectivity.

As for surjectivity, pick a degree 0 divisor D. Again by our observation, $\dim \mathcal{O}_X(D + O) = \deg(D + O) = 1$. Take any nonzero section $f \in \mathcal{O}_X(D + O)$. $\operatorname{div}(f) \geq -D - O$. Since $\deg \operatorname{div}(f) = 0$ but $\deg(-D - O) = -1$, $\operatorname{div} f = -D - O + p$ for some $p \in X$. Thus, $D = p - O + \operatorname{div}(-f)$, i.e. it is the image of p up to a principal divisor. ☐

§11. Bundle Properties

Acknowledgment. *This section is wholly taken from [Ati57].*

We lay out the tools needed for the bundle classification.

Remark 11.1. *Let us run through all the notation.*

- $\pi : E \twoheadrightarrow X$ *denotes a rank* $r < \infty$ *algebraic bundle over an elliptic curve* $(X, O, \mathcal{O}_X)$*.*
- $\Gamma(E)$ *is shorthand for* $\Gamma(X, \mathcal{O}_X(E))$*.*
- *The fiber* E_x *over* $x \in X$ *is isomorphic to* $\mathcal{O}_X(E)_x / \mathfrak{m}_x \mathcal{O}_X(E)_x$ *where* $\mathfrak{m}_x$ *is the maximal ideal of* $(\mathcal{O}_X)_x$*; hence* E_x *is a* $(\mathcal{O}_X)_x / \mathfrak{m}_x (\mathcal{O}_X)_x = k[X]_p / \mathfrak{m}_p$*-vector space.*
- $E(D) = E \otimes_{\mathcal{O}_X} \mathcal{O}_X(D)$ *and* $\mathcal{O}_X(n) = \mathcal{O}_X(nO)$*. We sometimes drop the* X *from* $\mathcal{O}_X$ *too.*
- *We often write* $H^q(X, \mathcal{F})$ *as just* $H^q(\mathcal{F})$*.*
- $\mathcal{E}(r, d)(X)$ *is the set of indecomposable rank* r*, degree* d *bundles on* X*.*

Definition 11.2 (Sufficient). E *has* <u>*sufficient sections*</u> *if* $\Gamma(E) \to E_x$ *is onto for all* $x \in X$*.*

We invoke this next theorem without proof.

Theorem 11.3 ([Ser56]). *For big enough* n*,* $E(n)$ *has sufficient sections and* $H^{q>0}(E(n)) = 0$*.*

Definition 11.4 (Trivial bundle). *A generic bundle* (E, X, F, p) *is* <u>*trivial*</u> *if* $E \cong X \times F$*.*

Proposition 11.5. *E has sufficient sections if and only if it is a quotient of a trivial bundle.*

Remark 11.6. *Pick a nonzero section $\phi \in \Gamma(E)$. The image of ϕ is a projective subvariety in E since ϕ is regular. At any $p \in X$, if t uniformizes $k[X]_p$, then $\phi = t^{v_x}\phi'$ for $v_x \geq 0, \phi'(p) \neq 0$. The collection of one-dimensional vector spaces $k[\phi'(x)] \subseteq E$ make-up a line bundle $[\phi] \subseteq E$, of which ϕ is a section.*

Define $\operatorname{div}(\phi) = \sum_{x \in X} v_x x$ *and* $\deg(\phi) = \sum_x v_x$ *so* $\deg \phi = \deg[\phi]$.

Corollary 11.7. *By the proposition, $E' = E/[\phi]$ has sufficient sections if E does.*

Lemma 11.8. *For any bundle E, we can find a sequence*

$$0 = E_0 \subsetneq E_1 = [\phi] \subsetneq \cdots \subsetneq E_r = E,$$

where $L_i = E_i/E_{i-1}$ is a line bundle.

Proof. $E' = E/[\phi]$ will have enough sections so simply repeat the process described in the remark above to produce the sequence. This will terminate at E_r since E is of dimension r.

By the aforementioned theorem, $E(n)$ has sufficient sections for large n. Thus,

$$0 = E_0(n) \subsetneq E_1(n) = [\phi] \otimes \mathcal{O}_X(n) \subsetneq \cdots \subsetneq E_r(n) = E(n), \qquad (\star)$$

for line bundles $L_i' = E_i(n)/E_{i-1}(n)$. But, the exact sequence

$$0 \to E_{i-1}(n) \to E_i(n) \to (E_i/E_{i-1}) \otimes \mathcal{O}_X(n) \to 0 \quad \text{shows} \quad L_i' \cong (E_i/E_{i-1})(n).$$

Thus $L_i = L_i'(-n)$ are line bundles so we are done after tensoring ($\star$) with $\mathcal{O}_X(-n)$. ☐

Definition 11.9 (Splitting). *A splitting of a bundle E is a tuple $(L_1, \ldots, L_r)$ described above.*

Lemma 11.10. *The integers $\deg \phi$ for nonzero $\phi \in \Gamma(E)$ are bounded above.*

Proof. Fix a splitting $0 = E_0 \subsetneq \cdots \subsetneq E_r = E$ with $L_i = E_i/E_{i-1}$.

For any nonzero $\phi \in \Gamma(E)$, there is some $i \geq 1$ so that $[\phi] \nsubseteq E_{i-1}$ but $[\phi] \subseteq E_i$. So $[\phi] \to L_i$ is nonzero. Thus $0 \neq \Gamma(\operatorname{Hom}([\phi], L_i)) \cong \Gamma([\phi]^\vee \otimes L_i)$ which happens only if $\deg([\phi]^\vee \otimes L_i) = \deg[\phi]^\vee + \deg L_i = \deg L_i - \deg \phi \geq 0$, i.e. $\deg L_i \geq \deg \phi$. Thus, $\deg \phi \leq \sup_i \deg L_i$. ☐

Since $\sup_i \deg L_i$ works for all ϕ, we can make the following definitions.

Definition 11.11 (Maximal). *A section $\phi \in \Gamma(E)$ is maximal if ϕ has maximal degree. $[\phi]$ is called a maximal line bundle of E.*

Definition 11.12 (Maximal splitting). *$(L_1, \ldots, L_r)$ is a maximal splitting of E if:*

(i) *L_1 is a maximal line-bundle,* (ii) *$(L_2, \ldots, L_r)$ is a maximal splitting of E/L_1.*

Remark 11.13. *If E has sufficient sections, a maximal splitting is possible.*

Lemma 11.14. *Say (L_1, L_2) is a maximal splitting for E. Then $\deg L_2 - \deg L_1 \leq 2g$.*

Proof. The exact sequence $0 \to \mathcal{O}_X \to E \otimes L_1^\vee \to L_2 \otimes L_1^\vee \to 0$ induces the long exact sequence:

$$0 \to H^0(\mathcal{O}_X) \to H^0(E \otimes L_1^\vee) \to H^0(E \otimes L_1^\vee) \to H^1(X, \mathcal{O}_X) \to \cdots$$

Suppose $\deg(L_2 \otimes L_1^\vee) \geq 2g+1$. then Riemann-Roch implies $\dim H^0(L_2 \otimes L_1^\vee) \geq (2g+1) - g + 1 = g + 2$. Note $\dim H^0(\mathcal{O}_X) = 1$ (the constants) and $\dim H^1(\mathcal{O}_X) = -\deg \mathcal{O}_X + g - 1 + \dim H^0(\mathcal{O}_X) = g$ so by exactness,

$$\dim H^0(E \otimes L_1^\vee) = -0 + \dim H^0(\mathcal{O}_X) + \dim H^0(L_2 \otimes L_1^\vee) - \dim H^1(\mathcal{O}_X) \geq 1 + g + 2 - g = 3.$$

Hence the kernel of $H^0(E \otimes L_1^\vee) \to (E \otimes L_1^\vee)_x$ has dimension at least one, implying the existence of a section $\phi \in \Gamma(E \otimes L_1^\vee)$ with $\operatorname{div}\phi \geq x$ and $\deg \phi \geq 1$.

Observe $[\phi] \otimes L_1 \subseteq E \otimes L_1 \otimes L_1^\vee \cong E$ yet $\deg([\phi] \otimes L_1) \geq 1 + \deg L_1$, contradicting the fact L_1 has maximal degree. So $\deg L_2 - \deg L_1 \leq 2g$. $\square$

Corollary 11.15. *Say $(L_1, \ldots, L_r)$ is a maximal splitting for E. Then $\deg L_i - \deg L_{i-1} \leq 2g$.*

Proof. Obvious by induction. $\square$

There are bounds in the opposite direction too.

Lemma 11.16. *Say E is indecomposable with sufficient sections. Then E has a maximal splitting $(L_1, \ldots, L_r)$ with $\deg L_i \geq \deg L_1 - (i-1)(2g-2)$.*

Proof. See: [Ati57]. $\square$

Corollary 11.17. *If X is elliptic, E is indecomposable, and $\Gamma(E) \neq 0$, then E has a maximal splitting $E = (L_1, \ldots, L_r)$ so that there are nontrivial maps L_1 into L_i for all i and $\mathcal{O}_X$ into L_1.*

Lemma 11.18. *If $E \in \mathcal{E}(r, d)$ with $0 \leq d < r$, then E has a trivial sub-bundle $I_s = I_{\dim \Gamma(E)}$.*

Proof. Consider a maximal splitting $E = (L_1, \ldots, L_r)$ ensured by the corollary. If $\deg L_1 > 0$,

$$d = \deg E = \sum \deg L_i \geq \sum \deg L_1 \geq r.$$

But $d < r$ by assumption so $\deg L_1 = 0$. Since $\mathcal{O}_X(0)$ maps nontrivially into L_1, $L_1 = \mathcal{O}_X(0)$.

However, L_1 is maximal (of maximal degree) so $\operatorname{div}\phi = 0$ for all $\phi \in \Gamma(E)$, i.e. $\Gamma(E) \hookrightarrow E_x$ for all $x \in X$ so $\Gamma(E)$ generates a trivial sub-bundle $I_{\dim \Gamma(E)} \subseteq E$. $\square$

Lemma 11.19. *If $E \in \mathcal{E}(r, r)$, then E has a max splitting $(L, \ldots, L)$ with $\deg L = 1$.*

Proof. By Riemann-Roch for elliptic curves, $\dim \Gamma(E) \geq \deg E = r$.

Take the maximal splitting $(L_1, \ldots, L_r)$ from the above corollary. If $\deg L_0 = 0$, then E has a trivial sub-bundle of $I_{\dim \Gamma(E)}$ of E so $\dim \Gamma(E) = r$ and $E = I_{\dim \Gamma(E)}$, but E is indecomposable - contradiction. So $\deg L_1 \geq 1$. However:

$$r = \deg E = \sum \deg L_i \geq r \deg L_1 \implies \deg L_i = 1 \text{ for all } i.$$

Thus, $L = L_1 \cong L_i$ with $\deg L = \deg L_i$ for all i so $E = (L, \ldots, L)$ suffices. $\square$

We need one last concept.

Definition 11.20 (Extension). *E is an extension of E' by E'' if*

$$0 \to E' \to E \to E'' \to 0$$

is exact. Such E are classified by $H^1(Hom(E'', E'))$ up to isomorphism.

Definition 11.21 (E', E''-partial, -complete). *Consider the following bundle diagram:*

$$\begin{tikzcd}
 & E_1 \neq 0 \arrow[d, tail, "j_1"] & & 0 \neq E_2 \arrow[dl, "j_2"'] & \\
0 \arrow[r] & E' \arrow[r, "i"] & E \arrow[ul, "q_1"'] \arrow[r, "p"] & E'' \arrow[u, two heads, "q_2"'] \arrow[r] & 0
\end{tikzcd}$$

An extension E is E'-partial if there exists (E_1, q_1, j_1) such that $q_1 i j_1 = 1_{E_1}$ and E''-partial if there exists (E_2, q_1, j_1) such that $q_2 p j_2 = 1_{E_2}$.

E is E' (E'')-complete if it is not *E' (E'')-partial.*

Lemma 11.22. *$0 \to E' \to E \to I_s \to 0$ is I_s-complete if and only if $p_* : \Gamma(E) \to \Gamma(I_s)$ is zero.*

Proof. If E is not I_s-complete, then we get a factor, say 1, of I_s taking the role of E_2 above. Hence $q_{2*} p_* j_{2*} = 1_{1*}$ so $p_* \neq 0$.

Conversely, if $p_* \neq 0$, there is $\phi \in \Gamma(E)$ so $p_*\phi = \psi \neq 0$. $\mathrm{div}(\psi) \geq \mathrm{div}(\phi) \geq 0$ but $\mathrm{div}(\psi) = 0$ since ψ is a section of a trivial bundle forcing $\mathrm{div}(\phi) = 0$ meaning $[\phi]$ is a line sub-bundle of E and $[\phi] \cong [\psi]$. $E \xrightarrow{p} I_s \xrightarrow{\text{proj}} [\psi] \to 0$ and $[\psi] \cong [\phi] \hookrightarrow E$ makes this I_s-partial, so not I_s-complete. □

Proposition 11.23. (i). *The classes of extensions $0 \to E' \to E \to I_s \to 0$ are in (1-1) correspondence with coboundary group maps $\delta : H^0(I_s) \to H^1(E')$.*

(ii). *E is I_s-complete iff δ is injective.*

(iii). *If $s = \dim H^1(E')$, there is a unique extension E that is I_s-complete.*

Proof. (i). We take this for granted [Ati57].

(ii). Use the cohomological long exact sequence of $E' \hookrightarrow E \twoheadrightarrow I_s$ with the last lemma.

(iii). Observe $s = \dim H^1(E') = \dim H^0(I_s)$. By (ii), E is I_s-complete if and only if δ is injective, i.e. an isomorphism. Because any two isomorphisms δ_1, δ_2 differ by precomposotion of an automorphism of $\Gamma(I_s)$, the I_s-complete extensions E_1, E_2 are isomorphic. □

Remark 11.24. *We can dualize an extension*

$$0 \to I_s \to E \to E' \to 0 \quad \rightsquigarrow \quad 0 \to E'^{\vee} \to E^{\vee} \to I_s^{\vee} \to 0$$

and apply the lemma to $\delta : H^1(I_s^{\vee}) \to H^1(E'^{\vee})$.

Although past results work for projective varieties, this next one requires X to be elliptic.

Lemma 11.25. *Say $0 \to I_s \to E \to E' \to 0$ is I_s-complete. Then $\dim \Gamma(E) = \dim \Gamma(E')$.*

Proof. Following the remark, we have the sequence

$$0 \to \Gamma(E'^{\vee}) \to \Gamma(E^{\vee}) \to \Gamma(I_s^{\vee}) \to \mathcal{H}^1(E'^{\vee}) \to \cdots$$

Since E is an I_s-complete extension, the above lemma says $\delta : H^1(I_s^{\vee}) \to H^1(E'^{\vee})$ is injective implying $\Gamma(E^{\vee}) \to \Gamma(I_s^{\vee})$ is surjective so $\Gamma(E'^{\vee}) \cong \Gamma(E^{\vee})$. By Serre duality, $H^1(E' \otimes \Omega_X) \cong H^1(E \otimes \Omega_X)$. However, Ω_X is a line bundle of degree $2g-2 = 0$ so $\Omega_X \cong \mathcal{O}_X$ so $H^1(E') \cong H^1(E)$.

Note $\deg E = \deg E' + \deg I_s = \deg E'$. By Riemann-Roch, $\dim \Gamma(E) = \dim \Gamma(E')$. □

§12. Bundle classification

Acknowledgment. *This section, too, is inspired wholly by [Ati57].*

We will show $\mathcal{E}(r, d)$ is basically a copy of X, where X is an elliptic curve with base-point corresponding to a fixed degree one line bundle A (see: theorem 10.12).

First, a uniqueness theorem.

Theorem 12.1. *Pick $E \in \mathcal{E}(r, d \geq 0)$. Say $s = \dim \Gamma(E)$.*

(i). *If $d = 0$, $s = 0, 1$. If $d > 0$, $s = d$.*

(ii). *If $d < r$, E has a trivial sub-bundle I_s and $E' = E/I_s$ is indecomposable with $\dim \Gamma(E') = s$.*

Proof. (a). Say $d \geq r > 0$. By a lemma on maximal splittings, $H^1(E) = 0$ so Riemann-Roch implies $s = d$. Also by Riemann-Roch, $\dim \Gamma(E) \geq \deg E = d > 0$. But earlier we showed that this implies there is a trivial sub-bundle I_s for which

$$0 \to I_s \to E \to E' \to 0, \quad E \text{ indecomposable, } I_s\text{-complete.}$$

So by the last lemma of Bundle Properties, $s = \dim \Gamma(E') = \dim H^1(E'^{\vee})$ (using Serre duality and remembering that $\mathcal{O}_X \cong \Omega_X$ for elliptic curves).

By dimension and I_s-complete, this corresponds to an isomorphism $\delta : H^1(I_s^*) \to H^1(E'^{\vee})$. If E' is decomposable, then $\delta = \delta_1 \oplus \delta_2$ corresponding to extensions E_1', E_2' of $I_{\dim H^1(E_1'^{\vee})}, I_{\dim H^1(E_2'^{\vee})}$ by E_1', E_2', but direct summing these extensions makes E decomposable - contradiction.

(b). Now say $0 \leq d < r$. We induct over r. If $r = 1$, $d = 0$ so (i) and (ii) hold.

Say this property holds for $0 < r' < r$. Thus, $E' \in \mathcal{E}(r - s, d)$ satisfies (i) and (ii) but so does E since $\dim \Gamma(E) = \dim \Gamma(E')$ also by the last lemma of Bundle Properties. □

Now, the existence theorem.

Theorem 12.2. *Pick $E' \in \mathcal{E}(r', d)$ and set $s = d$ if $d > 0$ and 1 if $d \neq 0$. If $d = 0$, also suppose $\Gamma(E') \neq 0$. There exists a unique bundle $E \in \mathcal{E}(r' + s, d)$ (up to isomorphism) given by an extension*

$$0 \to I_s \to E \to E' \to 0.$$

Proof. Part (i) of theorem 12.1 guarantees that $\dim \Gamma(E') = s$ so by the proposition from Bundle Properties, there is a unique I_s-complete extension E of E'. We win if E is indecomposable.

Suppose $E = F \oplus G$. $\dim \Gamma(E) = \dim \Gamma(E') = s$ so $\Gamma(I_s) \cong \Gamma(E)$ so F and G must contain trivial sub-bundles I_F, I_G with $I_F \oplus I_G = I_s$. Then $E' \cong E/I_s \cong F/I_F \oplus G/I_G$, but E' is indecomposable by assumption so without loss of generality $I_F = F$ but that would make E not I_s-complete - contradiction. So $E \in \mathcal{E}(r' + s, d)$. □

Theorem 12.3. (i). *For all r, there is a unique bundle $F_r \in \mathcal{E}(r, 0)$ with $\Gamma(F_r) \neq 0$ up to isomorphism. In fact,*

$$0 \to \mathcal{O}_x \to F_r \to F_{r-1} \to 0$$

(ii). *For $E \in \mathcal{E}(r, 0)$, $E \cong L \otimes F_r$ for a unique degree zero line bundle L.*

Proof. (i). Let $\overline{\mathcal{E}}(r, 0) = \{E \in \mathcal{E}(r, 0) : \Gamma(E) \neq 0\}/(E \cong F)$. We show $|\overline{\mathcal{E}}(r, 0)| = 1$ by induction.

It's well known that all degree zero line bundles are isomorphic so the base case $r = 1$ holds.

If the result holds for all $0 < r' < r$, then if $E \in \overline{\mathcal{E}}(r, 0)$, then by theorem 12.1 there is an exact sequence $0 \to 1 \to E \to E' \to 0$ for $E' \in \overline{\mathcal{E}}(r - 1, 0)$. By hypothesis, $E' \cong F_{r-1}$. Thus theorem 12.2 guarantees a unique bundle E satisfying

$$0 \to 1 \to E \to F_{r-1} \to 0$$

up to isomorphism, finishing this part of the proof.

(ii). If $E \in \mathcal{E}(r, 0)$, then $E \otimes A \in \mathcal{E}(r, r)$ so we get a splitting $E \otimes A = (L, L, \ldots, L)$ with $\deg L = 1$. So $E \otimes A \otimes L^\vee \cong F_r$, i.e. $E \cong F_r \otimes (A^\vee \otimes L)$ where $A^\vee \otimes L$ is degree $-1 + 1 = 0$ line bundle.

Since $F_r = (\mathcal{O}_X, \ldots, \mathcal{O}_X)$, $F_r \otimes A^\vee \otimes L = (A^\vee \otimes L, \ldots, A^\vee \otimes L)$. $If A^\vee \otimes L$ is not trivial, then $\dim \Gamma(F_r \otimes A^\vee \otimes L) \leq r \dim \Gamma(A^\vee \otimes L) = 0$, contradicting the fact $\dim \Gamma(F_r) = 1$ - so $F_r \cong F_r \otimes (A^\vee \otimes L)$ implies $A^\vee \otimes L \cong 1$, i.e. $A^\vee \otimes L$ is unique. □

Remark 12.4. *F_r has a trivial sub-bundle so $F_r^\vee$ also does. Thus, $\Gamma(F_r^\vee) \neq 0$ and $F_r^\vee \in \overline{\mathcal{E}}(r, 0)$.*

Theorem 12.5. *Fixed line bundle A determines a 1-1 correspondence $\mathcal{E}((r, d), 0) \longleftrightarrow \mathcal{E}(r, d)$.*

Proof. Observe that $-\otimes A$ already identifies $\mathcal{E}(r, d) \longleftrightarrow \mathcal{E}(r, d+r)$ so we can reduce to $0 \leq d < r$. If $d = 0$, we are done so suppose $0 < d < r$.

By part (ii) of theorem 12.1, we have an exact sequence $0 \to I_d \to E \to E' \to 0$ for $E' \in \mathcal{E}(r - d, d)$. And again theorem 12.2 ensures this E' came from a unique E, giving a 1-1 correspondence $\mathcal{E}(r - d, d) \longleftrightarrow \mathcal{E}(r, d)$.

Combining these two paragraphs shows, by Euclidean algorithm, that

$$\mathcal{E}((r, d), 0) \longleftrightarrow \mathcal{E}(r, d).$$

□

Remark 12.6. *Composing this last theorem with theorem 12.3 shows that $\mathcal{E}(r, d)$ is in 1-1 correspondence with $\mathcal{E}(1, 0)$. But degree 0, rank 1 line bundles precisely constitute the Picard variety $\mathrm{Pic}^0\ X$ which, by theorem 10.12, is in correspondence with X itself. This gives $\mathcal{E}(r, d)$ the structure of its base elliptic curve, concluding the classification.*

§13. References

[Ati57] M. F. Atiyah. "Vector Bundles Over an Elliptic Curve". In: *Proceedings of the London Mathematical Society* s3-7.1 (1957), pp. 414–452. DOI: https://doi.org/10.1112/plms/s3-7.1.414. eprint: https://londmathsoc.onlinelibrary.wiley.com/doi/pdf/10.1112/plms/s3-7.1.414. URL: https://londmathsoc.onlinelibrary.wiley.com/doi/abs/10.1112/plms/s3-7.1.414.

[AM94] M.F. Atiyah and I.G. MacDonald. *Introduction To Commutative Algebra*. Addison-Wesley series in mathematics. Avalon Publishing, 1994. ISBN: 9780813345444. URL: https://books.google.com/books?id=HOASFid4x18C.

[Don11] Simon Donaldson. *Riemann Surfaces*. Oxford Graduate Texts in Mathematics 22. Hardcover. 198 Madison Avenue, New York, NY 10016, United States of America: Oxford University Press, Jan. 1, 2011, pp. xiii, 286. ISBN: 978-0-19-852639-1. (Visited on 09/30/2023). published.

[Har77] Robin Hartshorne. *Algebraic Geometry*. Vol. 52. Graduate Texts in Mathematics. Springer, 1977. URL: http://www.worldcat.org/oclc/2798099.

[Lim20] Willie Rush Lim. *Riemann surfaces are always second-countable*. 2020. URL: https://willierushrush.github.io/posts/2020/05/second-countability/#fn4.

[Ser56] Jean-Pierre Serre. "Géométrie algébrique et géométrie analytique". fr. In: *Annales de l'Institut Fourier* 6 (1956), pp. 1–42. DOI: 10.5802/aif.59. URL: https://www.numdam.org/articles/10.5802/aif.59/.

[Sil86] Joseph H Silverman. *The Arithmetic of Elliptic Curves*. Vol. 106. Graduate Texts in Mathematics. Springer, 1986.

[Tel98] Constantin Teleman. *Riemann Surfaces*. Lecture notes, UC Berkeley. 1998. URL: https://math.berkeley.edu/~teleman/math/Riemann.pdf.

[WG08] Raymond O. Wells and O. García-Prada. *Differential Analysis on Complex Manifolds / by Raymond O. Wells*. eng. 3rd ed. 2008. Graduate Texts in Mathematics, 65. New York, NY: Springer New York, 2008. ISBN: 0-387-73892-4.